Physics

1. MECHANICS

1. Two steel balls fall freely on an elastic slab. The first ball is dropped from a height h_1 = 44 cm and the second from a height h_2 = 11 cm τ s after the first ball. After the passage of time τ, the velocities of the balls coincide in magnitude and direction.

   Determine the time τ and the time interval during which the velocities of the two balls will be equal, assuming that the balls do not collide.

2. A small ball moves at a constant velocity v along a horizontal surface and at point A falls into a vertical well of depth H and radius r. The velocity v of the ball forms an angle α with the diameter of the well drawn through point A (Fig. 1, top view).

   Determine the relation between v, H, r and α for which the ball can "get out" of the well after elastic impacts with the walls. Friction losses should be neglected.

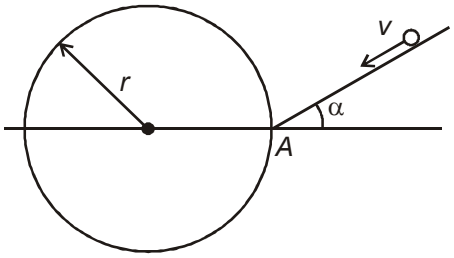


Fig. 1

3. A bobbin rolls without slipping over a horizontal surface so that the velocity v of the end of the thread (point A) is directed along the horizontal. A board hinged at point B leans against the bobbin (Fig. 2). The inner and outer radii of the bobbin are r and R respectively.

   Determine the angular velocity ω of the board as a function of an angle α.

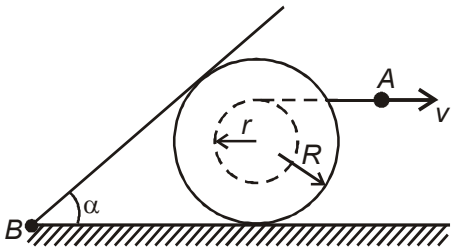


Fig. 2

4. A magnetic tape is wound on an empty spool rotating at a constant angular velocity. The final radius r_f of the winding was found to be three times as large as the initial radius r_i (Fig. 3). The winding time of the tape is t_1.

What is the time t_2 required for winding a tape whose thickness is half that of the initial tape?

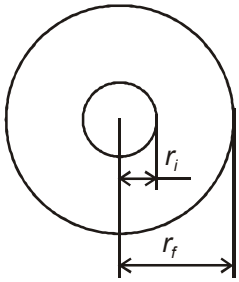


Fig. 3

5. Two rings O and O' are put on two vertical stationary rods AB and $A'B'$ respectively. An inextensible thread is fixed at point A' and on ring O and is passed through ring O' (Fig. 4).

   Assuming that ring O' moves downwards at a constant velocity v_1, determine the velocity v_2 of ring O if $\angle AOO' = \alpha$.

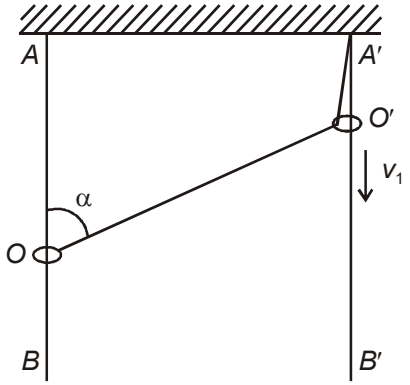


Fig. 4

6. The free end of a thread wound on a bobbin of inner radius r and outer radius R is passed round a nail A hammered into the wall (Fig. 5). The thread is pulled at a constant velocity v.

   Find the velocity v_0 of the centre of the bobbin at the instant when the thread forms an angle α with the vertical, assuming that the bobbin rolls over the horizontal surface without slipping.

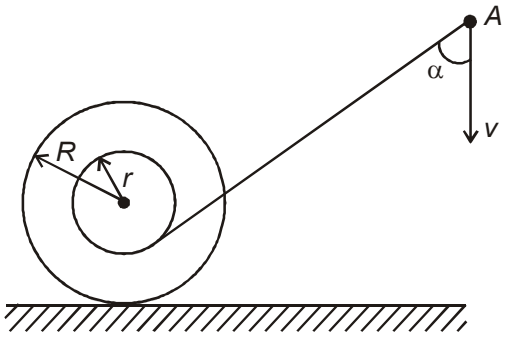


Fig. 5

7. A rigid ingot is pressed between two parallel guides moving in horizontal directions at opposite velocities v_1 and v_2.

   At a certain instant of time, the points of contact between the ingot and the guides lie on a straight line perpendicular to the directions of velocities v_1 and v_2 (Fig. 6).

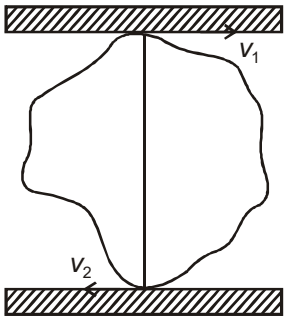


Fig. 6

What points of the ingot have velocities equal in magnitude to v_1 and v_2 at this instant?

8. Two identical weightless rods are hinged to each other and to a horizontal beam (Fig. 7). The rigidity of each rod is k_0 and the angle between them is 2α.

Determine the rigidity k of the system of rods relative to the vertical displacement of a hinge A under the action of a certain force F, assuming that displacements are small in comparison with the length of the rods.

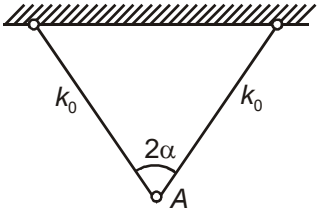


Fig. 7

9. A person hoists one of two loads of equal mass at a constant velocity v (Fig. 8). At the moment when the two loads are at the same height h, the upper pulley is released (is able to rotate without friction like the lower pulley).

Indicate the load which touches the floor first after a certain time t, assuming that the person continues to slack the rope at the same constant velocity v. The masses of the pulleys and the ropes and the elongation of the ropes should be neglected.

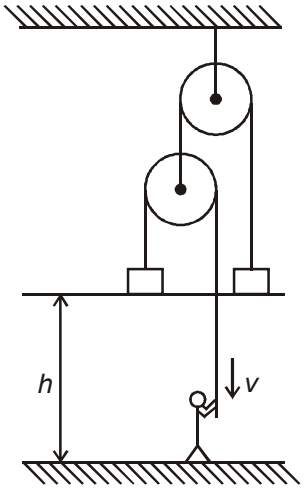


Fig. 8

10. Two balls are placed as shown in Fig. 9 on a "weightless" support formed by two smooth inclined planes each of which forms an angle α with the horizontal. The support can slide without friction along a horizontal plane. The upper ball of mass m_1 is released.

Determine the condition under which the lower ball of mass m_2 starts "climbing" up the support.

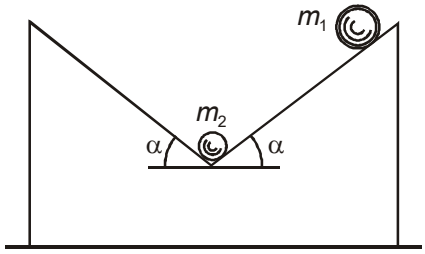


Fig. 9

11. A cylinder of mass *m* and radius *r* rests on two supports of the same height (Fig. 10). One support is stationary, while the other slides from under the cylinder at a velocity v.

Determine the force of normal pressure *N* exerted by the cylinder on the stationary support at the moment when the distance between points *A* and *B* of the supports is $AB = r\sqrt{2}$, assuming that the supports were very close to each other at the initial instant. The friction between the cylinder and the supports should be neglected.

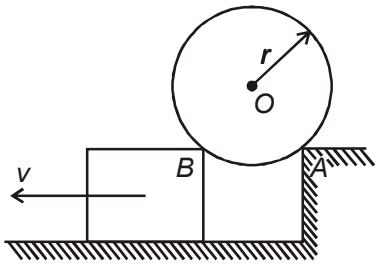


Fig. 10

12. A homogeneous rod *AB* of mass *m* and length ℓ leans with its lower end against the wall and is kept in the inclined position by a string *DC* (Fig. 11). The string is tied at point *C* to the wall and at point *D* to the rod so that $AD = \frac{AB}{3}$. The angles formed by the string and the rod with the wall are α and β respectively.

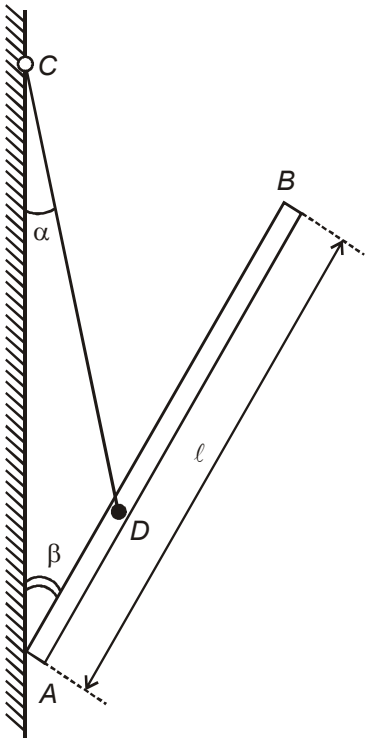


Fig. 11

Find all possible values of the coefficient of friction μ between the rod and the wall.

13. A massive disc rotates about a vertical axis at an angular velocity Ω. A smaller disc of mass m and radius r, whose axis is strictly vertical, is lowered on the first disc (Fig. 12). The distance between the axes of the discs is d ($d > r$), and the coefficient of friction between them is μ

Determine the steady-state angular velocity ω of the smaller disc. What moment of force M must be applied to the axis of the larger disc to maintain its velocity of rotation constant? The radius of the larger disc is $R > d + r$. The friction at the axes of the discs should be neglected.

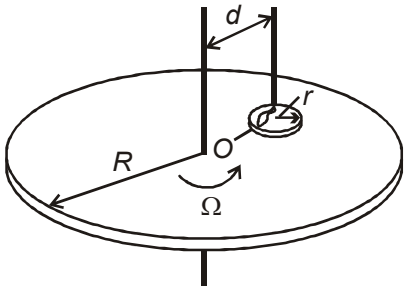


Fig. 12

14. A round box of inner diameter D containing a washer of radius r lies on a table (Fig. 13). The box is moved as a whole at a constant velocity v directed along the lines of centres of the box and the washer. At an instant t_0, the washer hits the box.

Determine the time dependences of the displacement x_{wash} of the washer and of its velocity v_{wash} relative to the table, starting from the instant t_0 and assuming that all the impacts of the washer against the box are perfectly elastic. Plot the graphs $x_{wash}(t)$ and $v_{wash}(t)$. The friction between the box and the washer should be neglected.

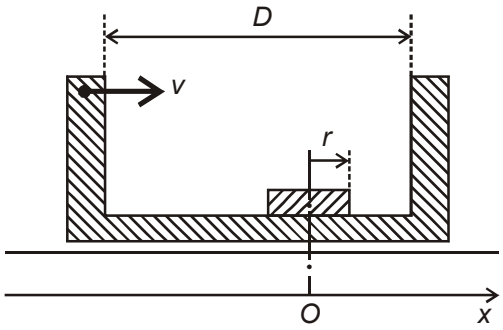


Fig. 13

15. A ring of mass m connecting freely two identical thin hoops of mass M each starts sliding down. The hoops move apart over a rough horizontal surface.

Determine the acceleration a of the ring at the initial instant if $\angle AO_1O_2 = \alpha$ (Fig. 14), neglecting the friction between the ring and the hoops.

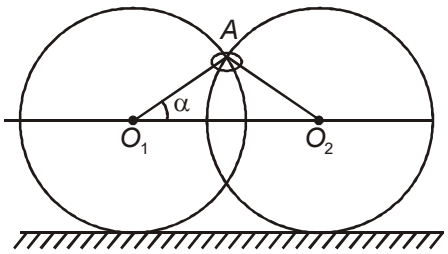


Fig. 14

16. A smooth washer impinges at a velocity v on a group of three smooth identical blocks resting on a smooth

horizontal surface as shown in Fig. 15. The mass of each block is equal to the mass of the washer. The diameter of the washer and its height are equal to the edge of the block.

Determine the velocities of all the bodies after the impact.

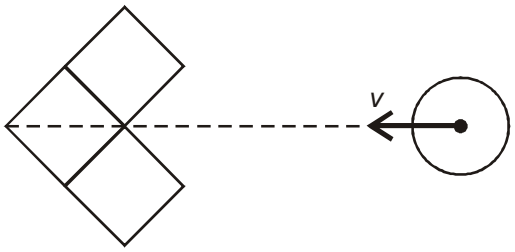


Fig. 15

17. A long smooth cylindrical pipe of radius r is tilted at an angle α to the horizontal (Fig. 16). A small body at point A is pushed upwards along the inner surface of the pipe so that the direction of its initial velocity forms an angle ϕ with generatrix AB.

    Determine the minimum initial velocity v_0 at which the body starts moving upwards without being separated from the surface of the pipe.

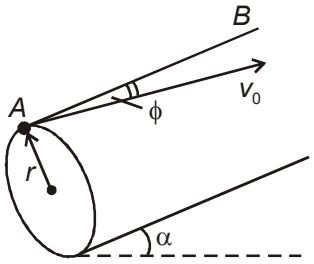


Fig. 16

18. An inextensible rope tied to the axle of a wheel of mass m and radius r is pulled in the horizontal direction in the plane of the wheel. The wheel rolls without jumping over a grid consisting of parallel horizontal rods arranged at a distance ℓ from one another ($\ell << r$).

    Determine the average tension T of the rope at which the wheel moves at a constant velocity v, assuming the mass of the wheel to be concentrated at its axle.

19. Two coupled wheels (i.e. light wheels of radius r fixed to a thin heavy axle) roll without slipping at a velocity v perpendicular to the boundary over a rough horizontal plane changing into an inclined plane of slope a (Fig. 17).

    Determine the value of v at which the coupled wheels roll from the horizontal to the inclined plane without being separated from the surface.

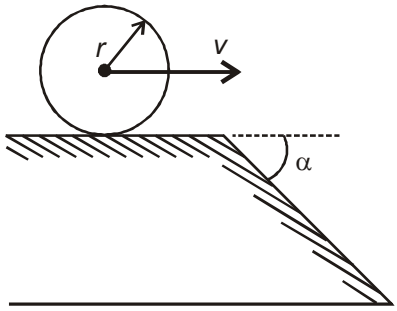


Fig. 17

20. A ball is tied by a weightless inextensible thread to a fixed cylinder of radius r. At the initial moment, the thread is wound so that the ball touches the cylinder. Then the ball acquires a velocity v in the radial direction, and the thread starts unwinding (Fig. 18).

Determine the length ℓ of the unwound segment of the thread by the instant of time t, neglecting the force of gravity.

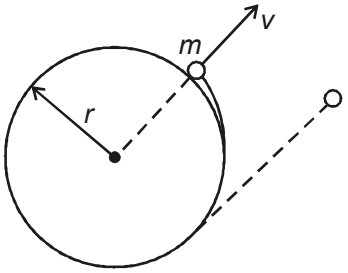


Fig. 18

21. A block is connected to an identical block through a weightless pulley by a weightless inextensible thread of length 2ℓ (Fig. 19). The left block rests on a table at a distance ℓ from its edge, while the right block is kept at the same level so that the thread is unstretched and does not sag, and then released.

    What will happen first: will the left block reach the edge of the table (and touch the pulley) or the right block hit the table?

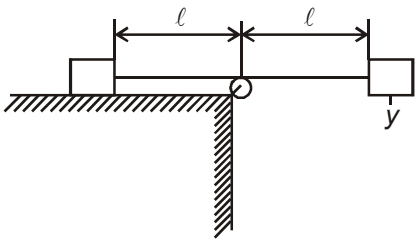


Fig. 19

22. Two loads of the same mass are tied to the ends of a weightless inextensible thread passed through a weightless pulley (Fig. 20). Initially, the system is at rest, and the loads are at the same level. Then the right load abruptly acquires a horizontal velocity v in the plane of the figure. Which load will be lower in a time?

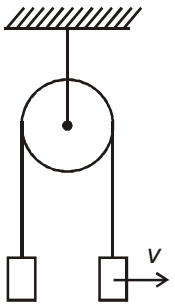


Fig. 20

23. Two balls of mass m_1 = 56 g and m_2 = 28 g are suspended on two threads of length ℓ_1 = 7 cm and ℓ_2 = 11 cm at the end of a freely hanging rod (Fig. 21).

    Determine the angular velocity ω at which the rod should be rotated about the vertical axle so that it remains in the vertical position.

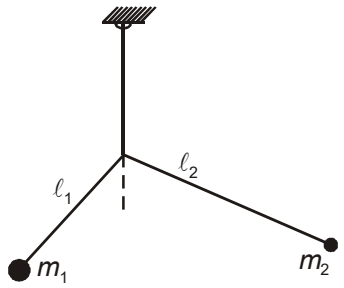


Fig. 21

24. Determine the minimum coefficient of friction μ_{min} between a thin homogeneous rod and a floor at which a person can slowly lift the rod from the floor without slippage to the vertical position, applying to its end a force perpendicular to it.

25. A thin perfectly rigid weightless rod with a point-like ball fixed at one end is deflected through a small angle α from its equilibrium position and then released. At the moment when the rod forms an angle $\beta < \alpha$ with the vertical, the ball undergoes a perfectly elastic collision with an inclined wall (Fig. 22).

    Determine the ratio $\frac{T_1}{T}$ of the period of oscillations of this pendulum to the period of oscillations of a simple pendulum having the same length.

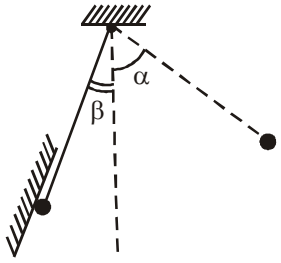


Fig. 22

26. Four weightless rods of length ℓ each are connected by hinged joints and form a rhomb (Fig. 23). A hinge *A* is fixed, and a load is suspended to a hinge *C*. Hinges *D* and *B* are connected by a weightless spring of length 1.5ℓ in the undeformed state. In equilibrium, the rods form angles $\alpha_0 = 30°$ with the vertical.

    Determine the period *T* of small oscillations of the load.

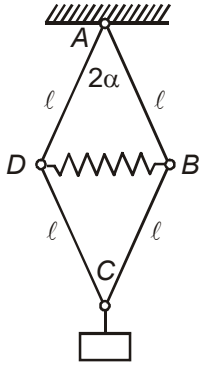


Fig. 23

27. A weightless rigid rod with a load at the end is hinged at point *A* to the wall so that it can rotate in all directions (Fig. 24).

    The rod is kept in the horizontal position by a vertical inextensible thread of length ℓ, fixed at its midpoint.

The load receives a momentum in the direction perpendicular to the plane of the figure.

Determine the period T of small oscillations of the system.

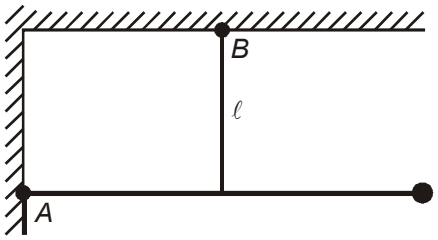


Fig. 24

28. One rope of a swing is fixed above the other rope by b. The distance between the poles of the swing is a. The lengths ℓ_1 and ℓ_2 of the ropes are such that $\ell_1^2 + \ell_2^2 = a^2 + b^2$ (Fig. 25).

Determine the period T of small oscillations of the swing, neglecting the height of the swinging person in comparison with the above lengths.

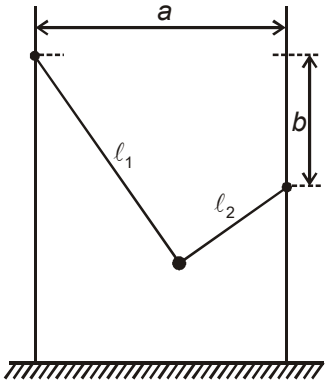


Fig. 25

29. Two cylinders with a horizontal and a vertical axis respectively rest on a horizontal surface. The cylinders are connected at the lower parts through a thin tube. The "horizontal" cylinder of radius r is open at one end and has a piston in it (Fig. 26). The "vertical" cylinder is open at the top. The cylinders contain water which completely fills the part of the horizontal cylinder behind the piston and is at a certain level in the vertical cylinder.

Determine the level h of water in the vertical cylinder at which the piston is in equilibrium, neglecting friction.

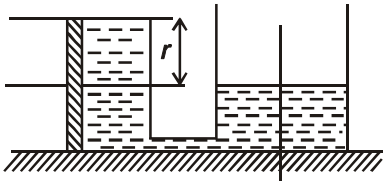


Fig. 26

30. A simple accelerometer (an instrument for measuring acceleration) can be made in the form of a tube filled with a liquid and bent as shown in Fig. 27. During motion, the level of the liquid in the left arm will be at a height h_1, and in the right arm at a height h_2.

Determine the acceleration a of a carriage in which the accelerometer is installed, assuming that the diameter of the tube is much smaller than h_1 and h_2.

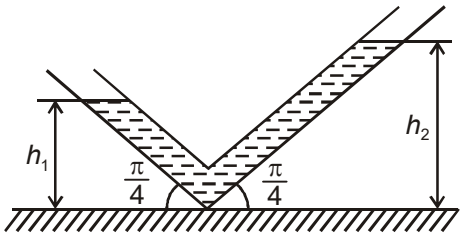


Fig. 27

31. Let us suppose that the drag F to the motion of a body in some medium depends on the velocity v of the body as $F = \mu v^{\alpha}$, where $\alpha > 0$.

At what values of the exponent α, will the body pass an infinitely large distance after an initial momentum has been imparted to it?

2. HEAT AND MOLECULAR PHYSICS

1. A cyclic process (cycle) 1-2-3-4-1 consisting of two isobars 2-3 and 4-1, isochor 1-2, and a certain process 3-4 represented by a straight line on the p-V diagram (Fig. 1) involves n moles of an ideal gas. The gas temperatures in states 1, 2 and 3 are T_1, T_2 and T_3 respectively and points 2 and 4 lie on the same isotherm.

Determine the work A done by the gas during the cycle.

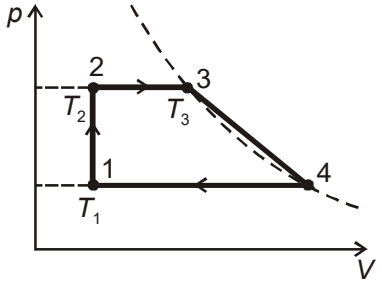


Fig. 1

2. A gas takes part in two thermal processes in which it is heated from the same initial state to the same final temperature.

The processes are shown on the p-V diagram by straight lines 1-3 and 1-2 (Fig. 2). Indicate the process in which the amount of heat supplied to the gas is larger.

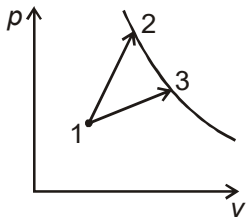


Fig. 2

3. A horizontal cylindrical vessel of length 2ℓ is separated by a thin heat-insulating piston into two equal parts each of which contains n moles of an ideal monatomic gas at a temperature T. The piston is connected to the end faces of the vessel by undeformed springs of rigidity k each (Fig. 4).

When an amount of heat Q is supplied to the gas in the right part, the piston is displaced to the left by a distance $x = \frac{\ell}{2}$.

Determine the amount of heat Q' given away at the temperature T to a thermostat with which the gas in

the left part is in thermal contact all the time.

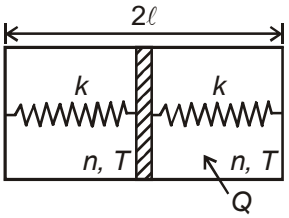


Fig. 4

4. A thermally insulated vessel is divided into two parts by a heat-insulating piston which can move in the vessel without friction. The left part of the vessel contains one mole of an ideal monatomic gas and the right part is empty. The piston is connected to the right wall of the vessel through a spring whose length in free state is equal to the length of the vessel (Fig. 5).

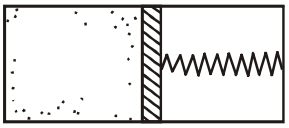


Fig. 5

Determine the heat capacity C of the system, neglecting the heat capacities of the vessel, piston and spring.

5. A thin-walled cylinder of mass m, height h, and cross-sectional area S is filled with a gas and floats on the surface of water (Fig. 6). As a result of leakage from the lower part of the cylinder, the depth of its submergence has increased by Δh.

Determine the initial pressure p_1 of the gas in the cylinder if the atmospheric pressure is p_0 and the temperature remains unchanged.

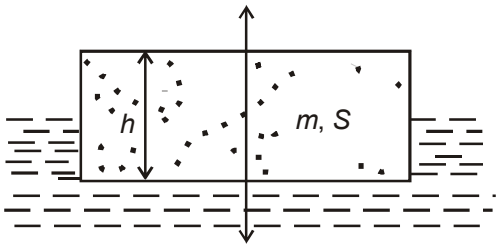


Fig. 6

6. A shock wave is the region of an elevated pressure propagating in the positive direction of the x-axis at a high velocity v.

At the moment of arrival of the wave, the pressure abruptly increases. This dependence is plotted in Fig. 7.

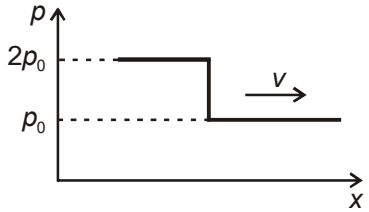


Fig. 7

Determine the velocity u acquired by a wedge immediately after the shock front passes through it. The mass of the wedge is m, and its size is shown in Fig. 8. Friction should be neglected, and the velocity acquired by the wedge should be assumed to be much lower than the velocity of the wave ($u << v$).

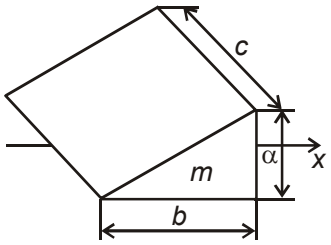


Fig. 8

3. ELECTRICITY AND MAGNETISM

1. The strength of the electric field produced by charges uniformly distributed over the surface of a hemisphere at its centre *O* is E_0. A part of the surface is isolated from this hemisphere by two planes passing through the same diameter and forming an angle α with each other (Fig. 1).

   Determine the electric field strength *E* produced at the same point *O* by the charges located on the isolated surface (on the "mericarp").

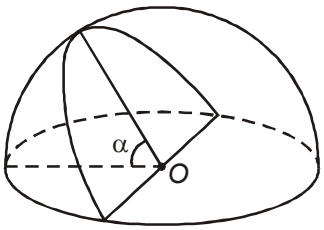


Fig. 1

2. A metal sphere having a radius r_1 charged to a potential ϕ_1 is enveloped by a thin-walled conducting spherical shell of radius r_2 (Fig. 2).

   Determine the potential ϕ_2 acquired by the sphere after it has been connected for a short time to the shell by a conductor.

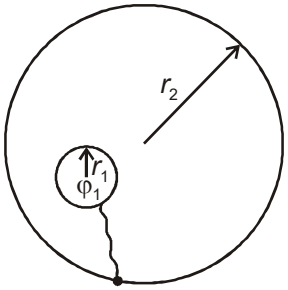


Fig. 2

3. A very small earthed conducting sphere is at a distance *a* from a point charge q_1 and at a distance *b* from a point charge q_2 ($a < b$). At a certain instant, the sphere starts expanding so that its radius grows according to the law $R = vt$.

   Determine the time dependence $I(t)$ of the current in the earthing conductor, assuming that the point charges and the centre of the sphere are at rest, and in due time the initial point charges get into the expanding sphere without touching it (through small holes).

4. A parallel-plate capacitor is filled by a dielectric whose permittivity varies with the applied voltage according to the law $\varepsilon = \alpha U$, where $\alpha = 1\ V^{-1}$. The same (but containing no dielectric) capacitor charged to a voltage $U_0 = 1.56$ V is connected in parallel to the first "nonlinear" uncharged capacitor.

   Determine the final voltage *U* across the capacitors.

5. Two small balls of mass m, bearing a charge q each, are connected by a nonconducting thread of length 2ℓ. At a certain instant, the middle of the thread starts moving at a constant velocity v perpendicular to the direction of the thread at the initial instant.

   Determine the minimum distance d between the balls.

6. The resistance of each resistor in the circuit diagram shown in Fig. 3 is the same and equal to R. The voltage across the terminals is U.

   Determine the current I in the leads if their resistance can be neglected.

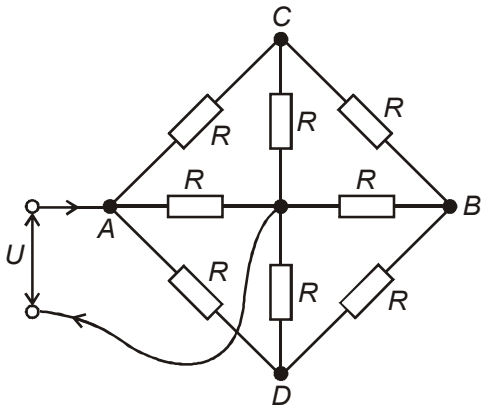


Fig. 3

7. Determine the resistance R_{AB} between points A and B of the frame formed by nine identical wires of resistance R each (Fig. 4).

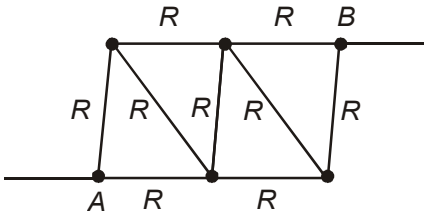


Fig. 4

8. The key K in circuit diagram shown in Fig. 5 can be either in position 1 or 2.

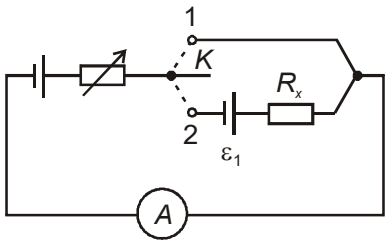


Fig. 5

   The circuit includes two d.c. sources, two resistors and an ammeter. The emf of one source is ε_1 and of the other is unknown. The internal resistance of the sources should be taken as zero. The resistance of the resistors is also unknown. One of the resistors has a varying resistance chosen in such a way that the current through the ammeter is the same for both positions of the key. The current is measured and is found to be equal to I.

   Determine the resistance denoted by R_x in the diagram.

9. The key K (Fig. 7) is connected in turn to each of the contacts over short identical time intervals so that the change in the charge on the capacitor over each connection is small.

   What will be the final charge q_f on the capacitor?

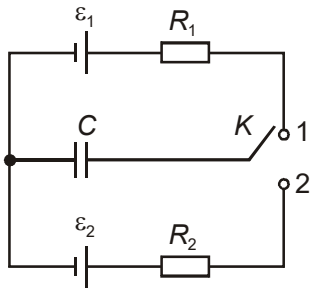


Fig. 7

10. A plate *A* of a parallel-plate capacitor is fixed, while a plate *B* is attached to the wall by a spring and can move, remaining parallel to the plate *A* (Fig. 9). After the key *K* is closed, the plate *B* starts moving and comes to rest in a new equilibrium position. The initial equilibrium separation *d* between the plates decreases in this case by 10%.

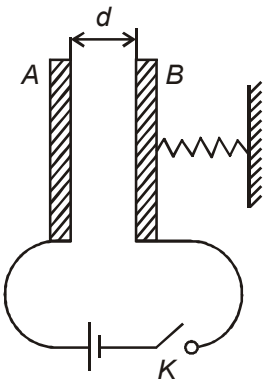


Fig. 9

What will be the decrease in the equilibrium separation between the plates if the key *K* is closed for such a short time that the plate *B* cannot be shifted noticeably?

11. An electric circuit consists of a current source of emf ε and internal resistance *r*, and two resistors connected in parallel to the source (Fig. 10). The resistance R_1 of one resistor remains unchanged, while the resistance R_2 of the other resistor can be chosen so that the power liberated in this resistor is maximum.

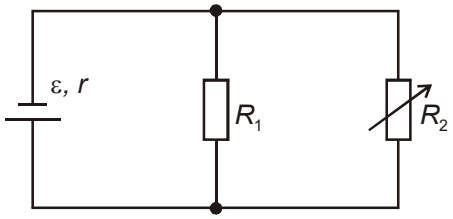


Fig. 10

Determine the value of R_2 corresponding to the maximum power.

12. A direct current flowing through the winding of a long cylindrical solenoid of radius *R* produces in it a uniform magnetic field of induction *B*. An electron flies into the solenoid along the radius between its turns (at right angles to the solenoid axis) at a velocity *v* (Fig. 11). After a certain time, the electron deflected by the magnetic field leaves the solenoid.

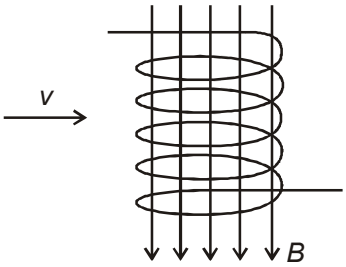


Fig. 11

Determine the time t during which the electron moves in the solenoid.

13. A quadratic undeformable superconducting loop of mass m and side a lies in a horizontal plane in a non-uniform magnetic field whose induction varies in space according to the law $B_x = -\alpha x$, $B_y = 0$, $B_z = \alpha z + B_0$ (Fig. 12). The inductance of the loop is L. At the initial moment, the centre of the loop coincides with the origin O and its sides are parallel to the x and y-axes. The current in the loop is zero and it is released.

How will it move and where will it be in time t after the beginning of motion?

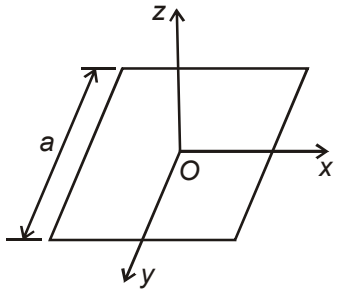


Fig. 12

14. A bank of two series-connected capacitors of capacitance C each is charged to a voltage U and is connected to a coil of inductance L so that an oscillatory circuit (Fig. 13) is formed at the initial moment. After a time τ, a breakdown occurs in one of the capacitors, and the resistance between its plates becomes zero.

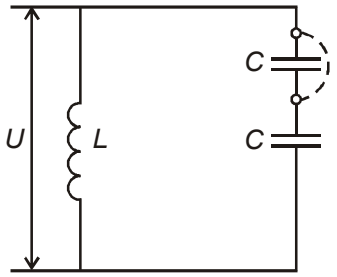


Fig. 13

Determine the amplitude q_0 of charge oscillations on the undamaged capacitor.

4. OPTICS

1. A point light source S is on the axis of a hollow cone with a mirror inner surface (Fig. 1). A converging lens produces on a screen the image of the source formed by the rays undergoing a single reflection at the inner surface (direct rays from the source do not fall on the lens).

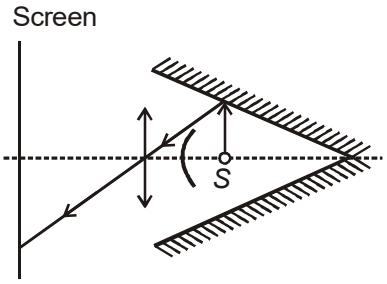


Fig. 1

What will happen to the image if the lens is covered by diaphragms like those shown in Figs. 2 and 3?

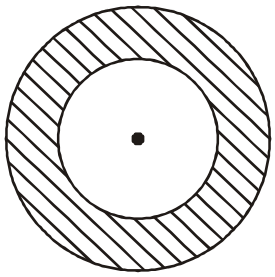


Fig. 2

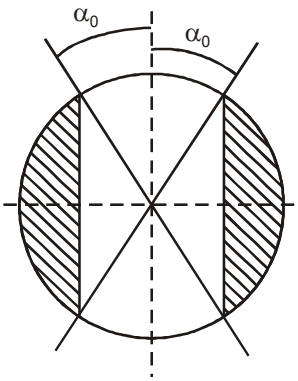


Fig. 3

2. The objective and the eyepiece of a telescope are double-convex symmetrical lenses made of glass with a refractive index n_{gl} = 1.5. The telescope is adjusted to infinity when the separation between the objective and the eyepiece is L_0 = 16 cm.

   Determine the distance L separating the objective and the eyepiece of the telescope adjusted to infinity with water poured in the space between the objective and the eyepiece (n_w = 1.3).

3. A fisherman lives on the shore of a bay forming a wedge with angle α in a house located at point A (Fig. 4). The distance from point A to the nearest point C of the bay is h and the distance from point D to roint A is ℓ. A friend of the fisherman lives across the bay in a house located at point B. Point B is symmetric (relative to the bay) to point A. The fisherman has a boat.

   Determine the minimum time t required for the fisherman to get to his friend from the house provided that he can move along the shore at a velocity v and row the boat at half that velocity (n = 2).

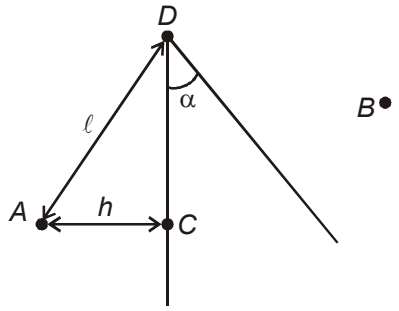


Fig. 4

4. The image of a point source S' lying at a distance b from a transparent sphere is formed by a small diaphragm only by rays close to the optical axis (Fig. 5).

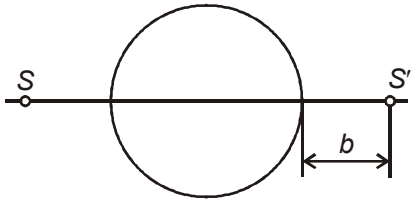


Fig. 5

Where will the image be after the sphere is cut into two parts perpendicular to the horizontal axis and the plane surface of the left half is silvered?

5. Let us suppose that a person seating opposite to you at the table wears glasses.

Can you determine whether he is shortsighted or long-sighted? Naturally, being a polite person, you would not ask him to let you try his glasses and in general would make no mention of them.

6. Two rays are incident on a spherical mirror of radius R = 5 cm parallel to its optical axis at distances h_1 = 0.5 cm and h_2 = 3 cm.

Determine the distance Δx between the points at which these rays intersect the optical axis after being reflected at the mirror.

Chemistry

1. An iron cylinder contains helium at a pressure of 250 kPa at 300 K. The cylinder can withstand a pressure of 1×10^6 Pa. The room in which cylinder is placed catches fire. Predict whether the cylinder will blow up before it melts or not, melting point of cylinder = 1800 K.
2. Two flasks of equal volume connected by a narrow tube (of negligible volume) are at 27°C and contains 0.70 mole of H_2 at 0.5 atm. One of the flask is then immersed into a bath kept at 127°C, while the other remains at 27°C. Calculate the final pressure and the number of mole of H_2 in each flask.
3. A 2 litre flask of N_2 at 20°C and 70 cm of Hg pressure is connected with a 3 litre of another flask of O_2 at the same temperature and 100 cm of Hg pressure. What will be the final pressure after the gases have throughly mixed at the same temperature as before? Also calculate the mole % of each gas in the resulting mixture. The volume of stop cock may be neglected.
4. Mercury diffusion pumps may be used in the laboratory to produce a high vacuum. Cold traps are generally placed between the pump and the system to be evacuated. These cause condensation of mercury vapours and prevent mercury from diffusing back into the system. The minimum pressure of mercury that can exist in the system is the vapour pressure of mercury at the temperature of cold trap. Calculate the number of mercury atom per unit volume in a cold trap at –120°C. The vapour pressure of mercury at this temperature is 10^{-16} torr.
5. A glass capillary tube sealed at both ends is 100 cm long. It lied horizontally with the middle 10 cm containing Hg. The two ends of the tube which are equal in length contain air at 27°C and pressure of 76 cm of Hg. The tube is kept in a horizontal position such that the air column at one end is at 0°C, the other end is maintained at 127°C. Calculate the length of the air column and its pressure which is at 0°C. Neglect the change in volume of Hg and glass.
6. A mixture of N_2 and water vapours is admitted to a flask which contains a solid drying agent. Immediately after admission, the pressure of the flask is 760 mm. After some hours, the pressure reached a steady value of 745 mm.
   (a) Calculate the composition in mole % of original mixture.
   (b) If the experiment is done at 20°C and the drying agent increase in mass by 0.15 g, what is the volume of flask?

   (Neglect volume occupied by drying agent)
7. A mixture of H_2O_v, CO_2 and N_2 was trapped in a glass apparatus with a volume of 0.731 mL. The pressure of total mixture was 1.74 mm of Hg at 23°C. The sample was transferred to a bulb in contact with dry ice (–75°C) so that H_2O_v are frozen out. When the sample returned to normal value of temperature, pressure was 1.32 mm of Hg. The sample was then transferred to a bulb in contact with liquid N_2 (–95°C) to freeze out CO_2. In the measured volume, pressure was 0.53 mm of Hg at original temperature. How many mole of each constituent are in mixture?
8. A mixture of N_2, NO and NO_2 in a closed container was analysed by selective absorption of the oxides of nitrogen. The initial pressure was 3.0 cm. After treatment with water, which absorb NO_2, the pressure left was 2.42 cm. A ferrous sulphate solution was then shaken with residual gas mixture to absorb NO after which the pressure was 1.24 cm. If vapour pressure of H_2O, in gaseous mixture after shaking with H_2O and in $FeSO_4$ solution were 0.2 cm and 0.18 cm respectively calculate the mole % of gases in initial mixture. Assume all measurements are made at same P and T.

9. A space capsule is filled with neon at 1 atm and 290 K. The gas effuses through a pin hole into outer space at the rate that pressure drops by 0.3 torr/sec. If the capsule is filled with 30 mole % of He, 20 mole % of O_2 and 50 mole % of N_2. What is the pressure drop under same condition?

10. Calculate the relative rates of diffusion for $^{235}UF_6$ and $^{238}UF_6$ in gaseous form. Also if naturally occurring uranium ore having U^{235} and U^{238} in the ratio 0.72 and 99.28% and if it is desired to enrich the U^{235} to 10% of the sample, making use of relative rates of diffusion of UF_6 having U^{235} and U^{238} isotopes, how many diffusion stages are required?

11. 'Bhopoon' an Indian toy for children producing sound operates on the principle that when pressure inside the chamber becomes equal to or more than operating pressure of valve called 'seete', then it produces sound. If the operating pressure of seete is 2 atm and chamber of 'Bhopoon' has volume of 300 mL, then how much minimum mass of air (in milligram) should be pumped to produce sound at 27°C?

    Assume R = 0.08 litre-atm mol^{-1} K^{-1} and molar mass of air = 29.0 g mol^{-1}.

12. A vacuum pump has a cylinder of volume v and is connected to a vessel of volume V to pump out air from the vessel. The initial pressure of gas in vessel is P. Show that after n strokes, the pressure in vessel is reduced to $P_n = P\left[\frac{V}{V+v}\right]^n$.

13. A graph is plotted for a van der Waals' gas between PV_m vs P leading to an intercept of 22.16 litre-atm. The temperature of gas at which these observations of P and V_M were made is ..... °C.

    (R = 0.08 litre atm K^{-1} mol^{-1})

14. A graph plotted between Pd vs P where P is pressure of C_2H_6 gas (assume ideal gas) and d is its density in g/L at particular temperature $\left[\frac{d\,Pd}{dP}\right] = 20$ when P = 8.21 atm, the temperature of gas will be :

    (a) 400 K (b) 1200 K

    (c) 300 K (d) 600 K

15. A mixture of SO_2 and O_2 in the molar ratio 16 : 1 is diffused through a pin hole for successive effusions three times to give a molar ratio 1 : 1 of diffused mixture. Which is/are correct if diffusion is made at same P and T in each operation?

    (a) Eight operations are needed to get 1 : 1 molar ratio

    (b) Rate of diffusion for SO_2 : O_2 after 8 operations is 0.707

    (c) Six operations are needed to get 2 : 1 molar ratio for SO_2 and O_2 in diffusion mixture

    (d) Rate of diffusion for SO_2 and O_2 after 6 operations is 2.41

16. Which of the following are correct statements?

    (a) Higher is $\frac{T_C}{P_C}$, larger is excluded volume

    (b) Higher is V_C, larger is excluded volume

    (c) Higher is $\frac{T_C}{V_C}$, more is van der Waals' constant for attraction

    (d) Smaller is size of molecule, lesser will be excluded volume

17. 50 mL of dry ammonia gas was sparked for a long time in an eudiometer tube over mercury. After sparking, the volume becomes 97 mL. After washing the gas with water and drying, the volume becomes 94 mL. This was mixed with 60.5 mL of oxygen and the mixture was burnt. After the completion of the combustion of H_2, the volume of the residual gas was 48.75 mL. Derive molecular formula of ammonia.

18. A sample of gaseous hydrocarbon occupying 1.12 litre at NTP, when completely burnt in air produced 2.2 g CO_2 and 1.8 g H_2O. Calculate the mass of hydrocarbon taken and the volume of O_2 at NTP required for its combustion.

19. A drop (0.05 mL) of 12 M HCl is spread over thin sheet of aluminium foil (thickness 0.10 mm and density of Al = 2.70 g/mL). Assuming whole of the HCl is used to dissolve Al, what will be the maximum area of hole produced in foil?

20. The molar mass of an organic acid was determined by the study of its barium salt. 4.290 g of salt was quantitatively converted to free acid by the reaction with 21.64 mL of 0.477 M H_2SO_4. The barium salt was found to have two mole of water of hydration per Ba^{2+} ion and the acid is monobasic. What is the molar mass of anhydrous acid?

21. For the reaction, $N_2O_5(g) \rightleftharpoons 2NO_2(g) + 0.5O_2(g)$, calculate the mole fraction of $N_2O_5(g)$ decomposed at a constant volume and temperature, if the initial pressure is 600 mm Hg and the pressure at equilibrium is 960 mm Hg. Assume ideal gas behaviour.

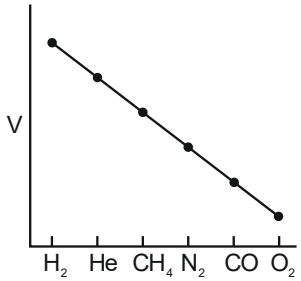
22. 25 mL of a solution of Na_2CO_3 having a specific gravity of 1.25 g mL^{-1} required 32.9 mL of a solution of HCl containing 109.5 g of the acid per litre for complete neutralization. Calculate the volume of 0.84 N H_2SO_4 that will be completely neutralized by 125 g of Na_2CO_3 solution.

23. A mixture of Xe and F_2 was heated and the white solid so formed reacted with H_2 to give 81 mL of Xe at STP and HF. The HF formed required 68.43 mL of 0.3172 M NaOH for complete neutralization. The empirical formula of white solid is XeF_n. The value of n is....

24. 23.1 mL of 0.115 M solution of NaOH is required for the end point using bromocresol green for 25 mL of 0.107 H_3PO_4. If the indicator phenolphthalein is used then 46.2 mL of same NaOH requires 25 mL of 0.107 M H_3PO_4. What is the ratio of replaceable H-atom in phenolphthalein and bromocresol indicators titrations?

25. A mixture of HCOOH and $H_2C_2O_4$ is heated with conc. H_2SO_4. The gaseous mixture on passing through KOH shows a reduction of $\frac{1}{6}$ in volume at same P and T. The mole ratio of HCOOH and $H_2C_2O_4$ in mixture is

26. Which of the following gases does not show correct variation in figure where volume vs gas plots are made? Each gas is taken in equal mass at STP.



(a) He and H_2

(b) CH_4 and He

(c) CO and N_2

(d) O_2 and CO

27. α-particles of 6 MeV energy is scattered back from a silver foil. Calculate the maximum volume in which the entire positive charge of the atom is supposed to be concentrated. (Z for silver = 47)

$K = 9.0 \times 10^9\ Nm^2C^{-2}$.

28. Energy required for excitation of electron in 1 mole H atom from ground state to 2nd excited state is 2.67 times lesser than dissociation energy per mole of $H_2(g)$. Calculate the amount of energy needed to excite each H atom of $H_2(g)$ confined in 1.0 litre at 27°C and 1 bar pressure.

R = 0.083 bar litre K^{-1} mol^{-1}; $R_H = 1.1 \times 10^7$ m^{-1}.

29. A hydrogen-like atom (atomic number Z) is in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successively emitting two photons of energies 10.20 eV and 17.00 eV respectively. Alternatively, the atom from the same excited state can make a transition to the second excited state by successively emitting two photons of energy 4.25 eV and 5.95 eV respectively. Determine the values of n and Z.

30. What is the maximum precision with which the momentum of an electron can be known if the uncertainty in the position of electron is ±0.001 Å? Will there be any problem in describing the momentum if it has a value of $\frac{h}{2\pi a_0}$, where a_0 is Bohr's radius of first orbit, i.e., 0.529 Å?

31. Wavelength of the K_α characteristic X-ray of iron and potassium are 1.931×10^{-8} and 3.737×10^{-8} cm respectively. What is the atomic number and name of the element for which characteristic K_α wavelength is 2.289×10^{-8} cm? [Use $\sqrt{v} = a(z-b)$]

32. 1.6 g sample of He^+ ions are excited to emit radiations. Spectral studies reveals that highest excited state is 2nd and total number of ions in 1st and 2nd excited state is one third the number of ions in ground state. Total energy evolved on de-excitation of sample is 429.5 kJ. Calculate the percentage of He^+ ions in 2nd excited state of original sample.

33. When photon of energy 4.25 eV strikes the surface, the ejected electron has maximum kinetic energy T_A expressed in eV and de Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated by another metal B by photons of 4.70 eV is T_B ($T_B = T_A - 1.5$ eV). If de Broglie wavelength of the electron is λ_B ($\lambda_B = 2\lambda_A$), then which is not correct?

(a) Work function of A is 2.25 eV (b) T_B = 0.5 eV

(c) Work function of B is 1.20 eV (d) T_A = 2.0 eV

34. The atomic masses of Li, He and proton are 7.01823 amu, 4.00387 amu and 1.00715 amu respectively. Calculate the energy evolved in the reaction,

$$_3Li^7 + {}_1p^1 \rightarrow 2\,{}_2He^4 + \Delta E$$

Given 1 amu = 931 MeV.

35. An isotopic species of lithium hydride $^6Li^2H$ is used as a potential nuclear fuel following the nuclear reaction:

$$^6_3Li\ ^2_1H \rightarrow 2\ ^4_2He$$

Calculate the expected power production of megawatt (MW) associated with 1.00 g of $^6Li\ ^2H$ per day assuming 100% efficiency. Given $^6_3Li = 6.01512$ amu; $^2_1H = 2.01410$ amu; $^4_2He = 4.00260$ amu.

36. A sample of $^{153}_{33}I$, as iodide ion, was administered to a patient in a carrier consisting of 0.10 mg of stable iodide ion. After 4 day, 67.7% of the initial activity was detected in the thyroid gland of the patient. What mass per cent of the stable iodide ion had migrated to the thyroid gland? $t_{\frac{1}{2}}$ for I = 8 day.

37. The radioactive disintegration of $_{94}Pu^{239}$ in an α-emission process is accompanied by the loss of 5.24 MeV/dis. If $t_{\frac{1}{2}}$ of $_{94}Pu^{239}$ is 2.44×10^4 year, calculate the energy released per year from 1.0 g sample of $_{94}Pu^{239}$ in kJ.

38. 1 g-atom of Ra^{226} is placed in an evacuated tube of volume 5 litre. Assuming that each $_{88}Ra^{226}$ nucleus is an α-emitter and all the contents are present in tube, calculate the total pressure of gases and partial pressure of He collected in tube at 27°C after the end of 800 year. $t_{\frac{1}{2}}$ of Ra is 1600 year. Neglect volume occupied by undecayed Ra.

39. In a sample of radioactive element, radium disintegrates at an average rate of 2.24×10^{23} α-particles per minute. Each α-particle takes up 2 electrons from the air and becomes a neutral helium atom. After 420 days, the He gas collected was 0.5 mL measured at 27°C and 750 nm of mercury pressure. From the above data, calculate Avogadro's no.

40. In an experiment on two radioactive isotopes of an element (which do not decay into each other), their molar ratio at a given instant is 3. The rapidly decaying isotope has larger mass and an initial activity of 1.0 μCi. The half-lives of the two isotopes are 12 and 16 hr respectively. What would be the activity of each isotope and their molar ratio after two day?

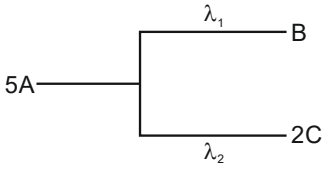
41. α-particle accelerated by 3×10^5 volt bombarded on a boron target. This results in the nuclear reaction.

$$_2He^4 + {}_5B^{10} \rightarrow {}_6C^{13} + {}_1H^1 + \gamma$$

If the combined energy of ^{13}C and H^1 is 5×10^5 eV, calculate the energy, frequency and wavelength of γ-rays. 1×10^5 eV energy is used in penetrating the nucleus.

(He = 4.0026 amu, B = 10.0129 amu, C = 13.0036 amu, H = 1.008 amu)

42. A follows parallel path 1st order reactions giving B and C as:



If initial concentration of A is 0.25 M, calculate the concentration of C after 5 hour of reaction.

Given, $\lambda_1 = 1.5 \times 10^{-5}\ s^{-1}$, $\lambda_2 = 5 \times 10^{-6}\ s^{-1}$

43. 0.209 g of radioactive Fe^{2+} is added with mass no. 57 to a solution of 1 litre which has 0.6 g of non-radioactive Fe^{3+} with mass no. 56 and the following reaction occurred.

$$^{57}Fe^{2+} + {}^{56}Fe^{3+} \rightarrow {}^{57}Fe^{3+} + {}^{56}Fe^{2+}$$

At the end of one hour, it was found that 10^{-5} moles of non-radioactive $^{56}Fe^{2+}$ was obtained and the rate of reaction was 3.38×10^{-7} mol l^{-1} hr^{-1}. Neglecting any change in volume, calculate the activity of the sample at the end of 1 hr. $t_{\frac{1}{2}}$ for $^{57}Fe^{2+}$ = 4.62 hr.

44. In a certain type of nuclear reaction, one neutron is a projectile (a reactant) and two neutrons are produced. Assume that each process takes 1 s. Suppose that half of all the product neutrons cause another event each, and the other half escape from the sample. How many neutrons will be produced in the third second?

45. A sample of radioactive element has rate R_1 at time t_1 and R_2 at time $t_2 (t_2 > t_1)$. Which one is not correct if λ is rate constant and τ is average life?

(a) $R_1 > R_2$

(b) No. of atoms decayed in time $(t_2 - t_1) = \dfrac{R_1 - R_2}{\lambda}$

(c) No. of atoms decayed in time $(t_2 - t_1) = (R_1 - R_2) \times \tau$

(d) No. of atoms decayed in time $(t_2 - t_1) = \dfrac{R_2 - R_1}{\lambda}$

46. Eight mole of chlorine (Cl_2) undergoes a loss and gain of 14 mole of electrons to form two oxidation state of chlorine (Cl). Write down the two half reaction and equation for disproportionation of chlorine, Cl_2.

47. Which of the following is not correct for the reaction?

$$(CN)_2 + 2OH^- \rightarrow CNO^- + CN^- + H_2O$$

(a) It is a disproportionation reaction

(b) N atom disproportionates and oxidation number of N are –3 in $(CN)_2$, –2 in CN^- and –5 in CNO^-

(c) C atom disproportionate and oxidation number of carbon are +3 in $(CN)_2$, +4 in CNO^- and +2 in CN^-

(d) $(CN)_2$ undergoes auto redox

48. Select the correct statements:

(a) Oxidation number of oxygen in O_2^+ is $+\frac{1}{2}$

(b) Oxidation number of oxygen in O_2^- is $-\frac{1}{2}$

(c) Oxidation number of Cr in K_3CrO_8 is +5

(d) Average oxidation number of Br in tribromooctaoxide (Br_3O_8) is $+\frac{18}{3}$

49. An ammeter and copper voltmeter are connected in series in an electric circuit through which a constant direct current flows. The ammeter shows 0.525 ampere. If 0.6354 g of Cu is deposited in one hour, what is percentage error of ammeter? Atomic mass of Cu= 63.54.

50. A test for complete removal of Cu^{2+} ions from a solution of Cu^{2+} (aq.) is to add NH_3 (aq.) A blue colour signifies the formation of complex $[Cu(NH_3)_4]^{2+}$ having $K_f = 1.1 \times 10^{13}$ and thus confirms the presence of Cu^{2+} in solution. 250 mL of 0.1 M $CuSO_4$ (aq.) is electrolysed by passing a current of 3.512 ampere for 1368 second. After passage of this charge sufficient quantity of NH_3 (aq.) is added to electrolysed solution maintaining $[NH_3]$ = 0.10 M. If $[Cu(NH_3)_4]^{2+}$ is detectable upto its concentration as low as 1×10^{-5}, would a blue colour be shown by the electrolysed solution on addition of NH_3?

51. In an electrolysis experiment, current was passed for 5 hour through two cells connected in series. The first cell contains a solution of gold and the second contains $CuSO_4$ solution. 9.85 g of gold was deposited in the first cell. If the oxidation no. of gold is +3, find the amount of Cu deposited on cathode in second cell. Also calculate the current strength in ampere. Atomic mass of Au = 197 and atomic mass of Cu = 63.5.

52. 50 mL of 0.1 M $CuSO_4$ solution is electrolysed using Pt electrodes with a current of 0.965 ampere for a period of 1 minute. Assuming that volume of solution does not change during electrolysis, calculate $[Cu^{2+}]$, $[H^+]$ and $[SO_4^{2-}]$ after electrolysis. What will be the concentration of each species if current is passed using Cu electrodes?

53. In the manufacture of Al, Al_2O_3 is dissolved in Na_3AlF_6 at 300 K and electrolysed between Al and carbon electrodes following the net reaction,

$$2Al_2O_3 \text{ (solution)} + 3C \rightarrow 4Al(l) + 3CO_2(g)$$

write the reaction of each electrode. Calculate the minimum voltage required between the electrodes if the Gibbs free energy change for the above reaction is –1370 kJ mol^{-1}.

54. During the discharge of a lead storage battery, the density of sulphuric acid fell from 1.294 g mL^{-1} to 1.139 g mL^{-1}. Sulphuric acid of density 1.294 g mL^{-1} is 39% by mass and that of density 1.139 g mL^{-1} is 20% by mass. The battery holds 3.5 litre of acid and the volume practically remained constant during the discharge. Calculate the no. of ampere hour for which the battery must have been used. The charging and discharging reactions are:

$$Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e \qquad \text{charging}$$

$$PbO_2 + 4H^+ + SO_4^{2-} + 2e \rightarrow PbSO_4 + 2H_2O \qquad \text{discharging}$$

55. If $NO_3^- \rightarrow NO_2$ (acidic medium); E° = 0.790 V and $NO_3^- \rightarrow NH_2OH$ (acidic medium); E° = 0.731 V. At what pH the above two half reactions will have same E values? Assume the concentrations of all the species other than H^+ to be unity.

56. Calculate E° of redox change : $Ag_2S + 2e \rightleftharpoons 2Ag + S^{2-}$ if the reaction occurs at pH = 3 and saturated with 0.1 M H_2S. K_1 and K_2 for H_2S are 1×10^{-8} and 1.1×10^{-13} respectively $K_{sp\ Ag_2S} = 2 \times 10^{-49}$ and $E^\circ_{Ag^+/Ag} = 0.8\ V$.

57. An alloy weighing 1.05 g of Pb — Ag was dissolved in desired amount of HNO_3 and the volume was made 350 mL. An Ag electrode was dipped in solution and E_{cell} of the cell $Pt\ \underset{1\,atm}{H_2}\ |H^+_{1M}||Ag^+|Ag$ was 0.503 V at 298 K. Calculate the percentage of lead in alloy. Given $E^\circ_{Ag^+/Ag} = 0.80\ V$.

58. For the reaction,

$$H_2(g) + 2AgCl(s) + 2H_2O(l) \rightarrow 2Ag(s) + 2H_3O^+(aq.) + 2Cl^-(aq.)$$

At 25°C, the standard free energy of formation of AgCl(s), $H_2O(l)$ and $(H_3O^+ + Cl^-)$ (aq.) are –109.7, –237.2 and –368.4 kJ/mol. Calculate what will be the cell voltage if this reaction runs at 25°C and one atmosphere in a cell in which H_2 activity is unity and H_3O^+(aq.) and Cl^-(aq.) activities are each at 0.01 M.

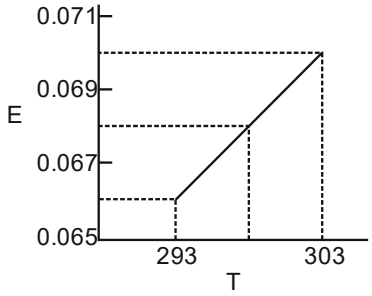
59. The overall formation constant for the reaction of 6 mole of CN^- with cobalt (II) is 1×10^{19}. Calculate the formation constant for the reaction of 6 mole of CN^- with cobalt (III). Given that,

$$Co(CN)_6^{3-} + e \rightarrow Co(CN)_6^{4-};\ E^\circ_{RP} = -0.83\ V$$

$$Co^{3+} + e \rightarrow Co^{2+};\ E^\circ_{RP} = 1.82\ V$$

60. For the cell:

Ag(s) | AgBr(s) | KBr(aq.) | Hg_2Br_2(s) | Hg(l), the variation of emf with temperature is shown by the graph given below:



(a) Write the cell reaction

(b) Calculate ΔG, ΔH and ΔS at 298 K

61. A current of 3 ampere was passed for 1 hour through an electrolyte solution of A_xB_y in water. If 2.977 g of A (atomic mass 106.4) was deposited at cathode and B was a monovalent ion, the formula of electrolyte was:

(a) AB_2

(b) AB

(c) AB_3

(d) AB_4

62. The E° for Cu^{2+}/Cu^+; Cu^+/Cu, Cu^{2+}/Cu are 0.15 V, 0.50 V and 0.325 V respectively. The redox cell showing redox reaction $2Cu^+ \rightarrow Cu^{2+} + Cu$ is made. The E° of this cell reaction and ΔG° may be:

(a) E° = 0.175 V or E° = 0.350 V

(b) n = 2 or 1 respectively

(c) ΔG° = –33.775 kJ

(d) All of the above

63. A half cell reaction : $Ag_2S_{(s)} + 2e \rightarrow 2Ag_{(s)} + S^{2-}$ is carried out in a half cell $Pt_{Ag_2S/Ag,\ H_2S}$ (0.1 M), at $[H^+] = 10^{-3}$. The emf of a half cell is:

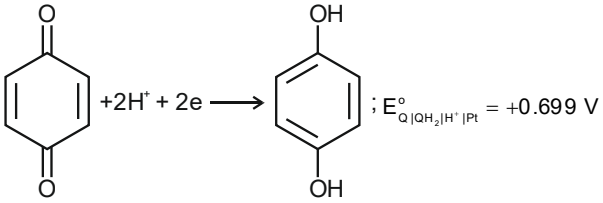
[if $E^\circ_{Ag^+/Ag} = 0.80$ V, $K_{a\,H_2S} = 10^{-21}$ and K_{sp} of $Ag_2S = 10^{-49}$]

(a) –0.1735 V (b) –0.19 V

(c) +0.1735 V (d) +0.19 V

64. A Quinhydrone electrode in contact of H^+ ion is coupled with standard calomel electrode. The E° of both electrodes are given as:

$$Pt\,|\,Q, QH_2\,|\,H^+ \,||\, 1M KCl\,|\,Hg_2Cl_2(s)\,|\,Hg(l)\,|\,Pt$$



$$\frac{1}{2}Hg_2Cl_2(s) + e \rightarrow Hg(l) + Cl^-;\ E^\circ_{Cl^-|Hg_2Cl_2|Hg} = +0.280\ V$$

If emf of cell so obtained is –0.124,, then pH is:

(a) 5 (b) 6

(c) 7 (d) 8

65. 25 mL of solution containing Fe^{2+} and Fe^{3+} sulphate acidified with H_2SO_4 is reduced by 3 g of metallic zinc. The solution required 34.25 mL of N/10 solution of $K_2Cr_2O_7$ for oxidation. Before reduction with zinc, 25 mL of the same solution required 22.45 mL of same $K_2Cr_2O_7$ solution. Calculate the strength of $FeSO_4$ and $Fe_2(SO_4)_3$ in solution.

66. Mg can reduce NO_3^- to NH_3 in basic solution:

$$NO_3^- + Mg(s) + H_2O \rightarrow Mg(OH)_2(s) + OH^-(aq.) + NH_3(g)$$

A 25.0 mL sample of NO_3^- solution was treated with Mg. The $NH_3(g)$ was passed into 50 mL of 0.15 N HCl. The excess HCl required 32.10 mL of 0.10 M NaOH for its neutralization. What was the molarity of NO_3^- ions in the original sample?

67. A sample of ferrous sulphate and ferrous oxalate was dissolved in dil. H_2SO_4. The complete oxidation of reaction mixture required 40 mL of N/15 $KMnO_4$. After the oxidation, the reaction mixture was reduced by Zn and H_2SO_4. On again oxidation by same $KMnO_4$, 25 mL were required. Calculate the ratio of Fe in ferrous sulphate and oxalate.

68. Hydroxylamine reduces iron III according to the equation:

$4Fe^{3+} + 2NH_2OH \rightarrow N_2O + H_2O + 4Fe^{2+} + 4H^+$. Iron II thus produced is estimated by titration with standard $KMnO_4$ solution. The reaction is

$$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

A 10 mL of hydroxylamine solution was diluted to one litre. 50 mL of this diluted solution was boiled with an excess of Fe^{3+} solution. The resulting solution required 12 mL of 0.02 M $KMnO_4$ solution for complete oxidation of Fe^{2+}. Calculate the mass of NH_2OH in one litre of original solution.

69. 25 mL of a solution of ferric alum $Fe_2(SO_4)_3 \cdot (NH_4)_2SO_4 \cdot 24H_2O$ containing 1.25 g of the salt was boiled with iron when the reaction $Fe + Fe_2(SO_4)_3 \rightarrow 3FeSO_4$ occurred. The unreacted iron was filtered off and solution is treated with 0.107 N $KMnO_4$ in acid medium. What is titre value? If Cu had been used in place of Fe, what would have been titre value?

70. For estimating ozone in the air, a certain volume of air is passed through an acidified or neutral KI solution when oxygen is evolved and iodide is oxidized to give iodine. When such a solution is acidified, free iodine is evolved which can be titrated with standard $Na_2S_2O_3$ solution. In an experiment 10 litre of air at 1 atm and 27°C were passed through an alkaline KI solution, at the end, the iodine entrapped in a solution on titration as above required 1.5 mL of 0.01N $Na_2S_2O_3$ solution. Calculate volume % of O_3 in sample.

71. A 1.7225 g of metal (bivalent) salt $A_x(CO_3)_y(OH)_z$ was dissolved to prepare 100 mL solution. 50 mL of this solution required 10 mL 1.0 N H_2SO_4 solution to reach the equivalence point using phenolphthalein as indicator. Another 50 mL solution using methyl orange as indicator required 15 mL of same acid. Deduce the formula of salt.

72. A 3.0 g sample of Cu_2O is dissolved in dil. H_2SO_4 where it undergoes disproportionation quantitatively. The solution is filtered off and 8.3 g pure KI crystals are added to clear filtrate in order to precipitate CuI with evolution of I_2. The solution is again filtered and boiled till all the I_2 is expelled. Now excess of an oxidizing agent is added to filtrate which liberates I_2 again. The liberated I_2 this time requires 10 mL of 1.0 N $Na_2S_2O_3$ solution. Calculate % by mass of Cu_2O in sample.

73. 1 g of moist sample of KCl and $KClO_3$ was dissolved in water to make 250 mL solution, 25 mL of this solution was treated with SO_2 to reduce chlorate to chloride and excess of SO_2 was removed by boiling. The total chloride was precipitated as silver chloride. The mass of precipitate was 0.1435 g. In another experiment, 25 mL of original solution was heated with 30 mL of 0.2 N ferrous sulphate solution and unreacted ferrous sulphate required 37.5 mL of 0.08 N solution of an oxidant for complete oxidation. Calculate the molar ratio of chlorate to chloride in the given mixture. Fe^{2+} reacts with ClO_3^- according to equation.

$$ClO_3^- + 6Fe^{2+} + 6H^+ \rightarrow Cl^- + 6Fe^{3+} + 3H_2O$$

74. Titration of I_2 produced from 0.1045 g of primary standard KIO_3 required 30.72 mL of sodium thiosulphate as shown below:

$$IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$$

$$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

The molarity of sodium thiosulphate ion is :

(a) 0.095 (b) 0.079

(c) 0.084 (d) 0.064

75. 2 mole, equimolar mixture of $Na_2C_2O_4$ and $H_2C_2O_4$ required V_1 L of 0.1 M $KMnO_4$ in acidic medium for complete oxidation. The same amount of the mixture required V_2 L of 0.1 M NaOH. The ratio of V_1 to V_2 is :

(a) 2 : 1 (b) 4 : 5

(c) 5 : 4 (d) 2 : 5

76. A mixture of $Na_2C_2O_4$ and $H_2C_2O_4$ requires 100 mL of 0.1 M $KMnO_4$ for complete neutralisation. The same mixture of neutralisation by a base requires 50 mL of 0.2 M NaOH solution. Which is/are correct?

(a) Mole ratio of $Na_2C_2O_4$ and $H_2C_2O_4$ = 4 : 1

(b) Equivalent ratio of $Na_2C_2O_4$ and $H_2C_2O_4$ = 4 : 1

(c) Mole of $C_2O_4^{2-}$ in mixture = 25×10^{-3}

(d) Mole ratio of $Na_2C_2O_4$ and $H_2C_2O_4$ = 1 : 4

77. A mixture of Cu_2S and CuS on oxidation by $Al(MnO_4)_3$ in acidic medium gives Cu^{2+} and SO_2. If molar masses of Cu_2S, CuS and $Al(MnO_4)_3$ are M_1, M_2 and M_3 respectively, therefore in this reaction which is/are correct?

(a) Equivalent mass of CuS is $\frac{M_1}{8}$ (b) Equivalent mass of CuS $\frac{M_2}{5}$

(c) Equivalent mass of $Al(MnO_4)_3$ is $\frac{M_3}{5}$ (d) Equivalent mass of $Al(MnO_4)_3$ is $\frac{M_3}{15}$

78. The vapour pressures of pure ethylene bromide ($C_2H_4Br_2$) and propylene bromide ($C_3H_6Br_2$) are 170 and 127 mm of Hg at a temperature. Find out the vapour pressure of ethylene bromide in a 60% by mass solution of ethylene bromide in propylene bromide at same temperature. Also report the vapour pressure of propylene bromide as well as total vapour pressure of solution.

79. Calculate the vapour pressure of solution having 3.42 g of cane-sugar in 180 g water at 40°C and 100°C. Given that boiling point of water is 100°C and heat of vaporisation is 10 kcal mol^{-1} in the given temperature range. Also calculate the lowering in vapour pressure of 0.2 molal cane-sugar at 40°C.

80. Two elements A and B form compounds having molecular formula AB_2 and AB_4. When dissolved in 20 g C_6H_6, 1 g of AB_2 lowers the f. pt. by 2.3°C whereas 1.0 g of AB_4 lowers it by 1.3°C. The K_f for C_6H_6 is 5.1 K mol^{-1} kg. Calculate atomic mass of A and B.

81. Two solutions of non-volatile solutes A and B having molar mass ratio of A and B (M_A/M_B) as 1/3 are prepared by dissolving 5% (mass/vol.) of each in water. Calculate the ratio of freezing point depression. If the two solutions are mixed to prepare a new solution (S_1) by mixing in the ratio 2 : 3 by volume and another new solution (S_2) is obtained by mixing in the ratio 3 : 2, find out $\frac{(\Delta T_f)_{s_1}}{(\Delta T_f)_{s_2}}$. Assume density of solution A and B equal to one.

82. A certain mass of a substance when dissolved in 100 g C_6H_6 lowers the freezing point by 1.28°C. The same mass of solute dissolved in 100 g of water lowers the freezing point by 1.40°C. If the substance has normal molar mass in benzene and is completely dissociated in water, into how many ions does it dissociate in water?

K_f' for H_2O and C_6H_6 are 1.86 and 5.12 K mol^{-1} kg respectively.

83. 1 g of a mixture containing NaCl and $CaCl_2$ is dissolved in water. Sodium oxalate on addition to this solution completely converts $CaCl_2$ to CaC_2O_4. The CaC_2O_4 is filtered and dissolved in dil. H_2SO_4. The clear solution requires 22 mL of 0.1 M $KMnO_4$. Calculate freezing point of an aqueous solution prepared by dissolving 5 g of same mixture containing $CaCl_2$ and NaCl in 100 g water. K_f of water is 1.86 K kg^{-1} mol^{-1}. Assume complete dissociation of $CaCl_2$ and NaCl.

84. The vapour pressure of a certain liquid is given by the equation:

$$\log_{10} P = 3.54595 + \frac{313.7}{T} + 1.40655 \log_{10} T,$$

where P is the vapour pressure in mm and T is temperature in K. Determine molar latent heat of vaporisation as a function of temperature and calculate its value at 80 K.

85. 100 g of 10% (mass/mass) urea solution is placed together with 200 g of 10% mass/mass glucose solution in closed jar to attain equilibrium. The number of mole of water transferred from one solution to other in order to attain equilibrium is ..... .

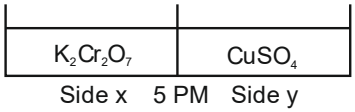
86. 18 g of liquid water is placed in an evacuated flask of 7.6 litre at 27°C. Vapour pressure of H_2O at 27°C is 24.63 mm of Hg. Assuming water vapour behaving as ideal gas and the volume of liquid water negligible in comparison to total volume of container, calculate the % of water that will be evaporated off in container.

87. 30 mL of 0.1 M KI_{aq} and 10 mL of 0.2 M $AgNO_3$ are mixed. The solution is then filtered out. Assuming no change in total volume, the resulting solution will freeze at : (Given, K_f for H_2O = 1.86 K kg mol^{-1}, assume molarity = molality)

(a) 0.28 (b) 0.22

(c) 0.0744 (d) 0.149

88. 0.05 M $CuSO_4$ when treated with 0.01 M $K_2Cr_2O_7$ gives green colour solution of $Cu_2Cr_2O_7$. The two solutions are separated as shown in figure. Due to osmosis



(a) Green colour formation inside Y (b) Green colour formation inside X

(c) Molarity of $K_2Cr_2O_2$ is lowered (d) Molarity of $CuSO_4$ solution is lowered

89. 2 litre of 1 molar solution of a complex salt $CrCl_3 \cdot 6H_2O$ (molar mass 266.5 g mol^{-1}) shows an osmotic pressure of 98.52 atm. The solution is now treated with 1 litre of 6 M $AgNO_3$, which of the following is/are correct?

(a) Mass of AgCl precipitated is 861 g

(b) The clear solution will show an osmotic pressure = 98.52 atm

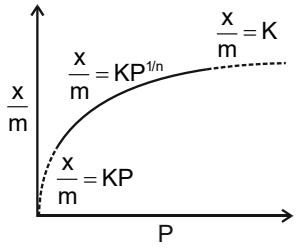
(c) The clear solution will show an osmotic pressure = 65.68 atm

(d) 2 mole of $[Cr(H_2O)_6]$ $(NO_3)_3$ will be present in solution

90. A solution of palmitic acid (molar mass = 256 g mol^{-1}) in benzene contains 4.24 g of acid per litre. This solution on pouring on water surface forms a monomolecular level of palmitic acid as benzene gets evaporated. If 500 cm^2 area of water surface is to be covered by a monolayer, what volume of solution in benzene is needed? Area covered by one molecule of palmitic acid is 0.21 nm^2.

91. The chemisorption of H_2 on an activated surface becomes 40% faster if temperature is raised from 500 K to 1000 K. Calculate energy of activation.

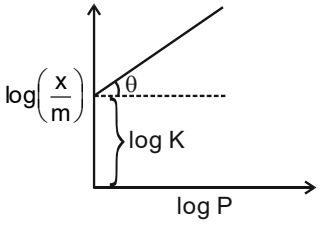
92. Which one is not correct about Freundlich isotherm? If



(a) $n = \frac{1}{\tan\theta}$ at average pressure

(b) θ = 45° at low pressure

(c) θ = 45° at high pressure

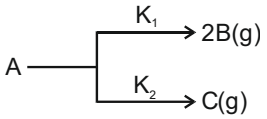
(d) 

93. A catalyst has total surface area of 10^3 cm^2 and adsorption sites available on it are 6×10^{15} sites/cm^2. H_2 gas is adsorbed on this surface occupying only 5% of the available sites of total surface area. On heating catalyst releases H_2 back and whole of the H_2 adsorbed is measured to occupy 2.46 mL at 0.03 atm and 300 K. The no. of H_2 molecules adsorbed on each site is equal to:

(a) 6 (b) 2

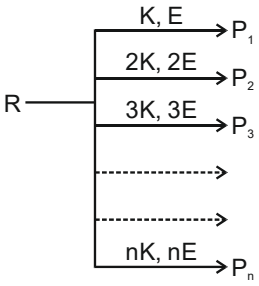
(c) 3 (d) 4

94. Gas A(g) shows two parallel path 1st order reaction as:



The initial pressure of A(g) in a container of volume V litre is 1 atm. After 10 s the pressure becomes 1.4 atm and after ∞ time it becomes 1.5 atm. Calculate K_1 and K_2.

95. A compound (R) undergoes parallel path reactions involving n steps as shown below:



Where K and E represents rate constant and energy of activations. Calculate the overall rate constant and overall energy of activation.

96. For the reaction, $2NO + H_2 \rightarrow N_2O + H_2O$, the value of $-\frac{dP}{dt}$ was found to be 1.5 Pa s^{-1} for a pressure of 359 Pa of NO and 0.25 Pa s^{-1} for a pressure of 152 Pa of NO. The pressure of H_2 being constant. If pressure of NO was kept constant, the value of $-\frac{dP}{dt}$ was found 1.60 Pa s^{-1} for a pressure of H_2 289 Pa and 0.79 Pa s^{-1} for a pressure of 147 Pa of H_2. Calculate the order of reaction.

97. The kinetics of hydrolysis of methyl acetate in excess dilute HCl at 25°C were followed by withdrawing 2 mL of the reaction mixture at intervals of (t), adding 50 mL water and titrating with baryta water. Determine the velocity constant of hydrolysis.

t (in minute)	0	75	119	259	∞
Titre value (in mL)	19.24	24.20	26.60	32.23	42.03

98. A vessel contains dimethyl ether at a pressure of 0.4 atm. Dimethyl ether decomposes as:

$CH_3OCH_3(g) \rightarrow CH_4(g) + CO(g) + H_2(g)$

The rate constant of decomposition is 4.78×10^{-3} min^{-1}, Calculate the ratio of initial rate of diffusion to rate of diffusion after 4.5 hour of initiation of decomposition. Assume composition of gas present and gas diffused remains same.

99. Decomposition of H_2O_2 is a first order reaction. A solution of H_2O_2 labelled as 20 volume was left open. Due to this some H_2O_2 decomposed. To determine the new volume strength after 6 hr, 10 mL of this solution was diluted to 100 mL. 10 mL of this diluted solution was titrated against 25 mL of 0.25 M $KMnO_4$ acidified solution. Calculate the rate constant for the decomposition of H_2O_2.

100. A certain reaction B^{n+} is getting converted to $B^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing agent which reacts only with B^{n+} and $B^{(n+4)+}$. In the process it converts B^{n+} to $B^{(n+2)+}$ and $B^{(n+4)+}$ to $B^{(n-1)+}$. At t = 0, the volume of reagent consumed is 25 mL and at t = 10 minute, the volume used is 32 mL. Calculate the rate constant for the conversion of B^{n+} of $B^{(n+4)+}$ assuming it to be a first order reaction.

101. Which one is incorrect statement?

(a) Always $E_a > \Delta H$ for an endothermic reaction

(b) $E_b \geqslant \Delta H$ for an endothermic reaction

(c) $E_a \geqslant \Delta H$ for an exothermic reaction

(d) Always $E_a < \Delta H$ for an exothermic reaction

102. For a gaseous reaction : $A_{(g)} \rightarrow B_{(g)}$, the rate expression may be given as:

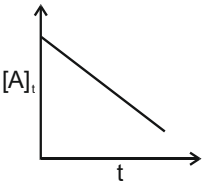
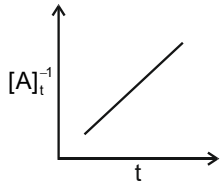
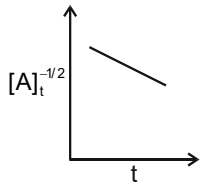
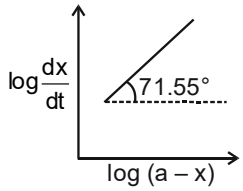
(a) $-\frac{d[A]}{dt} = K[A]^n$

(b) $-\frac{1}{V}\frac{dn_A}{dt} = K[A]^n$

(c) $-\frac{1}{RT}\frac{dP_A}{dt} = K[A]^n$

(d) $-\frac{dP}{dt} = K(P_A)^n$

103. Match the following Lists. (O.R. = order of reaction)

List A	List B
(a) 	(i) O.R. = 1/2
(b) 	(ii) O.R. = zero
(c) 	(iii) O.R. = 2
(d) 	(iv) O.R. = 3

104. In an equilibrium A + B $\rightleftharpoons$ C + D; A and B are mixed in a vessel at temperature T. The initial concentration of A was twice the initial concentration of B. After the equilibrium has reached, concentration of C was thrice the equilibrium concentration of B. Calculate K_C.

105. A mixture of SO_3, SO_2 and O_2 gases is maintained at equilibrium in 10 litre flask at a temperature at which K_C for the reaction,

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is 100 mol^{-1} litre. At equilibrium

(a) If no. of mole of SO_3 and SO_2 in flask are same, how many mole of O_2 are present?

(b) If no. of mole of SO_3 in flask are twice the no. of mole of SO_2, how many mole of O_2 are present?

106. At 90°C, the equilibrium, $H_2(g) + S(s) \rightleftharpoons H_2S(g)$ was established having $K = 6.8 \times 10^{-2}$. If 0.20 mole of hydrogen gas and 1.0 mole of sulphur are heated to 90°C in one litre vessel, what will be the partial pressure of H_2S at equilibrium?

107. The plot of $\log_{10} K_P$ against $\frac{1}{T}$ for the reaction:

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$$

is a straight line with slope 4.95×10^3. Calculate K_P at 25°C if standard entropies for $SO_2(g)$, $O_2(g)$ and $SO_3(g)$ are 248.2, 205.1 and 256.8 JK^{-1} mol^{-1} at 25°C respectively.

108. The standard enthalpy change and standard free energy change for the equilibrium:

$$Zn(s) + H_2O(g) \rightleftharpoons ZnO(s) + H_2(g)$$

are 224 kJ mol^{-1} and 33 kJ mol^{-1} respectively at the temperature 920 K. Calculate the temperature at which equilibrium constant becomes greater than 1. Assume ΔH remains constant.

109. When 1-pentyne (A) is treated with 4 N alcoholic KOH at 175°C, it is converted slowly into an equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne (B) and 3.5% of 1, 2-pentadiene (C). The equilibrium was maintained at 175°C. Calculate $\Delta G°$ for the following equilibria:

$B \rightleftharpoons A \quad ; \quad \Delta G_1^0 = ?$

$B \rightleftharpoons C \quad ; \quad \Delta G_2^0 = ?$

From the calculated value of ΔG_1^0 and ΔG_2^0 indicate the order of stability of (A), (B) and (C). Write a reasonable reaction mechanism showing all intermediates leading to (A), (B) and (C).

110. The amount of CO_2 in a gaseous mixture of CO_2 and CO is determined by passing the mixture into an aqueous solution that contains excess of $Ba(OH)_2$. The CO_2 reacts yielding a precipitate of $BaCO_3$ but CO does not react. This method is used to analyse the equilibrium composition of gas when 1.77 g CO_2 reacted with 2.0 g graphite in a 1.0 litre container at 1100 K. The analysis yielded 3.41 g of $BaCO_3$. Use this data to calculate K_P at 1100 K for the reaction.

$$CO_2(g) + C(s) \rightleftharpoons 2CO$$

111. In the preparation of quick lime from limestone, the reaction is,

$$CaCO_3 \rightleftharpoons CaO(s) + CO_2(g)$$

Experiments carried out between 850°C and 950°C led to set of K_P values fitting in empirical equation $\ln K_P = 7.282 - \frac{8500}{T}$ where T is absolute temperature. If the reaction is carried out in quiet air, what minimum temperature would be predicted from this equation for almost complete decomposition of lime?

112. 1 mole of A in 1 litre vessel maintained at constant T shows the equilibrium:

$A(g) \rightleftharpoons B(g) + 2C(g)$ K_{C_1}

$C(g) \rightleftharpoons 2D(g) + 3B(g)$ K_{C_2}

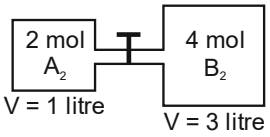
If the equilibrium pressure is $\frac{13}{6}$ times of initial pressure and $[C]_{eq.} = \frac{4}{9}[A]_{eq.}$, calculate K_{C_1} and K_{C_2}.

113. Gas A (mol. mass 16) and gas B (mol. mass 4) are confined into a vessel at partial pressure of 2 atm each at 27°C. The gases are then allowed to effuse out at 27°C through a value till the pressure inside the vessel becomes 2 atm after which the valve is closed and the gases are heated to 127°C and allowed to react as $A + B \rightleftharpoons C$, K_P = 0.1 atm^{-1}. Calculate the partial pressure of C at equilibrium assuming C is not formed till the effusing process is completed at 27°C.

114. A vessel of 250 litre containing 0.01 mole of Sb_2S_3 was filled with 0.01 mole of H_2 and reaction at 440°C,

$Sb_2S_3(s) + 3H_2(g) \rightleftharpoons 2Sb(s) + 3H_2S(g)$

was allowed to attain equilibrium. The H_2S formed was just sufficient to react with $Pb(NO_3)_2$ to give 1.19 g of PbS. The value of K_C at 440°C is ..... .

115. Gases are 

present in two containers at 300 K separated by a narrow tube of negligible volume having valve in between. On opening the valve the reaction $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ attains equilibrium at 300 K. If K_c = 4 at 300 K, the concentration of AB at equilibrium is:

(a) 1.33 M

(b) 2.66 M

(c) 1.66 M

(d) 0.66 M

116. Calculate the $[OH^-]$, $[NH_2 \cdot C_2H_4NH_3]^+$ and $[H_3N - C_2H_4NH_3]^{2+}$ in 0.15 M ethylene diamine (aq.); if

$NH_2C_2H_4NH_2 + H_2O \rightleftharpoons NH_2C_2H_4NH_3^+ + OH^-; K_1 = 8.5 \times 10^{-5}$

$NH_2C_2H_4NH_3^+ + H_2O \rightleftharpoons [NH_3C_2H_4NH_3]^{2+} + OH^-; K_2 = 2.7 \times 10^{-8}$

117. A solution of weak acid was titrated with base NaOH. The equivalence point was reached when 36.12 mL of 0.1 M NaOH have been added. Now 18.06 mL 0.1 M HCl were added to titrated solution, the pH was found to be 4.92. What is K_a of acid?

118. A solution contains a mixture of Ag^+ (0.10 M) and Hg_2^{2+} (0.10 M) which are to be separated by selective precipitation. Calculate the maximum concentration of iodide ion at which one of them gets precipitated almost completely. What % of that metal ion is precipitated?

(K_{sp} of AgI = 8.5 × 10^{-17} and K_{sp} of Hg_2I_2 = 2.5 × 10^{-26})

119. The solubility of $CaCO_3$ is 7 mg/litre. Calculate the solubility product of $BaCO_3$ from this information and from the fact that when Na_2CO_3 is added slowly to a solution containing equimolar concentration of Ca^{2+} and Ba^{2+}, no precipitate of $CaCO_3$ is formed until 90% of Ba^{2+} has been precipitated as $BaCO_3$.

120. H_2S is bubbled into a 0.02 M NaCN solution which is 0.02 M each in $Ag(CN)_2^-$ and $Cd(CN)_4^{-2}$. If K_{sp} of Ag_2S and K_{sp} of CdS are 1.0×10^{-50} and 7.1×10^{-28} and $K_{instability}$ for $[Ag(CN)_2^-]$ and $[Cd(CN)_4]^{2-}$ are 1.0×10^{-20} and 7.8×10^{-18}, which sulphide will precipitate first?

121. A 500 mL sample of an equilibrium mixture of gaseous N_2O_4 and NO_2 at 25°C and 753 mm of Hg was allowed to react with enough water to make 250.0 mL of solution at 25°C. Assume that all the dissolved N_2O_4 is converted to NO_2 which disproportionates in water yielding a solution of nitrous acid and nitric acid. Assume further that disproportionation reaction goes to completion and that none of the nitrous acid disproportionates. The equilibrium constant (K_p) for $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 0.113 at 25°C. K_a for HNO_2 is 4.5×10^{-4} at 25°C.

(a) Write balanced equation for disproportionation.

(b) What is molar concentration of NO_2^- and pH of the solution?

(c) What is osmotic pressure of solution?

(d) How many grams of lime (CaO) would be required to neutralize the solution?

122. A buffer solution of 0.080 M Na_2HPO_4 and 0.020 M Na_3PO_4 is prepared. The electrolytic oxidation of 1.0 m mole RNHOH is carried out in 100 mL buffer to give

$$RNHOH + H_2O \rightarrow RNO_2 + 4H^+ + 4e$$

Calculate approximate pH of the solution after oxidation is complete. pK_{a_1}, pK_{a_2} and pK_{a_3} of H_3PO_4 are 2.12, 7.20 and 12.0 respectively.

123. The pK_a of acetyl salicylic acid (aspirin) is 3.5. The pH of gastric juice in human stomach is about 2-3 and the pH in the small intestine is about 8. Aspirin will be:

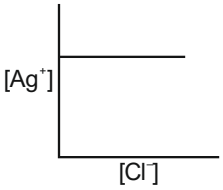
(a) Unionised in small intestine and in the stomach

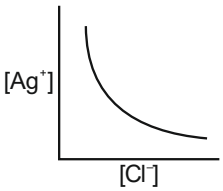
(b) Completely ionised in the small intestine and in the stomach

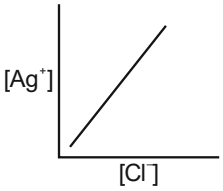
(c) Ionised in the stomach and almost unionised in the small intestine

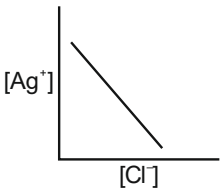
(d) Ionised in the small intestine and almost unionised in the stomach

124. In a saturated solution of AgCl, addition of NaCl is made drop by drop in excess. Which graph correctly represents the change?

(a) 

(b) 

(c) 

(d) 

125. The resistance of a solution 'A' is 50 ohm and that of solution 'B' is 100 ohm, both solutions being taken in the same conductivity cell. If equal volumes of solutions A and B are mixed, what will be the resistance of the mixture using the same cell? (Assume that there is no increase in the degree of dissociation of A and B on mixing)

126. A big irregular shaped vessel contained water, the conductivity of which was 2.56×10^{-5} S cm^{-1}. 500 g of NaCl was then added to the water and the conductivity after the addition of NaCl, was found to be 3.10×10^{-5} S cm^{-1}. Find the capacity of the vessel if it is filled with water (Λ_M° of NaCl = 149.9).

127. Conductivity of a saturated aqueous solution of $Co_2[Fe(CN)_6]$ is 2.06×10^{-6} S cm^{-1}. The water used to prepare the solution has conductivity 4.1×10^{-7} S cm^{-1}. If ionic molar conductivities of Co^{2+} and $[Fe(CN)_6]^{4-}$ are 86 S cm^2 mol^{-1} and 444.0 S cm^2 mol^{-2} respectively. Calculate the solubility product of $Co_2[Fe(CN)_6]$.

128. A water sample of swimming pool shows the resistance of 9.2×10^3 ohm at 25°C using a conductivity cell. The same cell dipped in 0.02 M KCl has the resistance of 85 ohm. Now 0.5 kg sodium chloride is dissolved in pool and throughly stirred. The solution sample now shows the resistance of 7.6×10^3 ohm. If Λ_M° for NaCl and Λ_v for 0.2 M KCl are 126.5 S cm^2 mol^{-1} and 138.0 S cm^2 mol^{-1}, calculate the volume of water in pool.

129. A graph plotted for log (i – 1) vs log Λ_v, where i is van't Hoff factor of given uni-univalent electrolyte having molar conductance Λ_v at dilution v, then which are correct?

(a) Slope = –log Λ° ; intercept = –1

(b) Slope = 1; intercept = – log Λ_m°

(c) Slope = 1; intercept = log Λ_m°

(d) Slope = log Λ_m° ; intercept = +1

130. An insulated container is divided into two equal portions. One portion contains an ideal gas at pressure P and temperature T. The other portion is a perfect vacuum. If a hole is opened between two portions, calculate:

(a) the change in internal energy

(b) the change in temperature of gas

131. Water is boiled under a pressure of 1.0 atm. When an electric current of 0.50 A from a 12V supply is passed for 300 s through a resistance in thermal contact with it, it is found that 0.789 g of water is vaporized. Calculate the molar internal energy and enthalpy changes at boiling point (373.15 K)

132. A flask of 1 litre having NH_3(g) at 2.0 atm and 200 K is connected with another flask of volume 800 mL having HCl(g) at 8 atm and 200 K through a narrow tube of negligible volume. The two gases reacts to form NH_4Cl(s) with evolution of 43 kJ mol^{-1} heat. If heat capacity of HCl(g) at constant volume is 20 JK^{-1} mol^{-1} and neglecting heat capacity of flask, NH_4Cl and volume of solid NH_4Cl formed, calculate the heat produced, final temperature and final pressure in flasks. (Assume R = 0.08 litre-atm K^{-1} mol^{-1}).

133. An athlete in a gymnasium room lifts a 50 kg mass through a vertical distance of 2.0 m; g = 9.8 ms^{-2}. The mass is allowed to fall through 2.0 m distance while coupled to an electrical generator. The electrical generator produces an equal amount of electrical work. This electrical work is used to produce aluminium by Hall's process involving the change.

$$Al_2O_3(\text{molten}) + 3C(s) \longrightarrow 2Al(l) + 3CO(g)$$

The reaction require standard free energy change equal to 593 kJ. How many times must the athlete lift the 50 kg mass and to drop and couple with generator to produce sufficient Gibbs free energy to produce 27 g Al ?

134. Calculate the entropy change when 1 kg of water is heated from 27°C to 200°C forming super heated steam under constant pressure. Given specific heat of water = 4180 J/kg -K and specific heat of steam = 1670 + 0.49 T J/kg -K and latent heat of vaporization = 23×10^5 J/kg

135. The standard enthalpy and entropy changes for the reaction in equilibrium for the forward direction are given below :

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

$\Delta H^{\circ}_{300K} = -41.16\ kJmol^{-1}$;

$\Delta S^{\circ}_{300K} = -4.24 \times 10^{-2}\ kJmol^{-1}$

$\Delta H^{\circ}_{1200K} = -32.93\ kJmol^{-1}$;

$\Delta S^{\circ}_{1200K} = -2.96 \times 10^{-2}\ kJmol^{-1}$

Calculate K_P at each temperature and predict the direction of reaction at 300 K and 1200 K, when $P_{CO} = P_{CO_2} = P_{H_2} = P_{H_2O} = 1$ atm at initial state.

136. Exactly 3.0 g of carbon was burned to CO_2 in copper calorimeter. The mass of calorimeter is 1.5 kg and the mass of water in calorimeter is 2 kg. The initial temperature was 20°C and final temperature was 31.5°C. If specific heat of copper is 0.0826 cal/g–K, the heat value to closes integer of carbon in kcal/g is .....

137. Two mole of an ideal monoatomic gas undergoes a reversible process for which PV^2 = constant. The gas expands from 1 litre to 3 litre starting from initial temperature of 300 K and the value of ΔH for the expansion in kcal is .....

138. If for the reaction at 300 K : $2Mg_{(s)} + O_{2(g)} \longrightarrow 2MgO_{(s)}$; $\Delta_r H = -1202\ kJ\ mol^{-1}$ and $\Delta_r S = -217.0\ JK^{-1}\ mol^{-1}$.

The total entropy change $(\Delta S)_T$ and Gibbs energy change during the course of reaction $(\Delta_r G)$ are respectively :

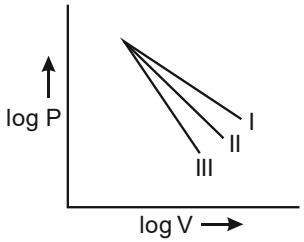
(a) 3.79×10^3 J, – 1136.9 kJ

(b) 3.79×10^3 J, + 1000 J

(c) +1000 J, 3.79×10^3 J

(d) –1136.9 kJ, 3.79×10^3 J

139. The following curves represents adiabatic expansions of gases He, O_2 and O_3 not necessarily in order. Which curve represents for O_3?



(a) I

(b) II

(c) III

(d) Any one of these

140. An ideal gas is allowed to expand adiabatically both reversibly and irreversibly in a closed system. If T_i is the initial temperature and T_f is the final temperature, which of the following statement is correct?

(a) $(T_f)_{irrev} > (T_f)_{rev}$

(b) $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process

(c) $(T_f)_{rev} = (T_f)_{irrev}$.

(d) $T_f = T_i$ for both reversible and irreversible process

141. The densities of graphite and diamond are 2 and 3 g/mL at 298 K respectively. If the $\Delta H - \Delta U$ for the change $C_G \rightarrow C_D$ is equal to 100 J mol^{-1}, the pressure at which graphite will be transformed into diamond is :

(a) 5×10^7 Pa (b) 9.92×10^5 Pa

(c) 9.92×10^{10} Pa (d) 9.92×10^8 Pa

142. Select the correct statement :

(a) There is a natural symmetry between converting work to heat and converting heat to work.

(b) No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.

(c) For every chemical reaction at equilibrium standard change in Gibbs free energy is zero.

(d) A constant temperature and pressure, chemical reactions are spontaneous in the direction of decreasing Gibbs energy.

143. Select the correct statements w.r.t. isothermal process:

(a) The magnitude of work involved in an intermediate irreversible expansion is less than that involved in reversible expansion

(b) Heat absorbed during reversible expansion is more than that in intermediate irreversible expansion

(c) The magnitude of work involved in an irreversible compression is more than that involved in intermediate irreversible compression

(d) Heat released during intermediate irreversible compression is more than that in reversible compression

144. For a reaction to be spontaneous in neither direction for a closed system:

(a) $(\Delta G)_{T, P} = 0$ (b) $(\Delta G)_{T, P} = -ve$

(c) $(\Delta S)_{U, V} = 0$ (d) $(\Delta S)_{U, V} = +ve$

145. Standard heat of formation of HgO(s) at 298 K and at constant pressure is –90.8 kJ mol^{-1}. If excess of HgO(s) absorbs 41.84 kJ of heat at constant volume, calculate the amount of Hg that can be obtained at constant volume and 298 K. (At wt. of Hg = 200).

146. A cooking gas cylinder is assumed to contain 11.2 kg isobutane having heat of combustion –2658 kJ mol^{-1}. If a family needs 15×10^3 kJ energy per day for cooking, how long a cylinder can last? Also if 30% heat is lost due to incomplete combustion, then how long would it take to last?

147. An athlete takes 20 breaths per minute at room temperature. The air inhaled in each breath is 200 mL which contains 20% oxygen by volume, while exhaled air contains 10% oxygen by volume. Assuming that all the oxygen consumed if used for converting glucose into CO_2 and $H_2O(l)$, how much glucose will be burnt in the body in one hour and what is the heat produced? (Room temperature = 27°C and enthalpy of combustion of glucose is –2822.5 kJ mol^{-1} at 0°C)

148. The temperature of a bomb calorimeter was found to rise by 1.617 K when a current of 3.20 ampere was passed for 3 seconds from a 12.0 V source. 100 g sample of a liquid (molar mass = 40) is now placed in this calorimeter and a passage of 0.5 ampere current for 3 minutes from a 1.50 V source gave a temperature rise of 0.8°C. What is molar heat capacity of liquid?

149. The enthalpies of neutralisation of a strong acid HA and weak acid HB by NaOH are –13.68 kcal/eq. and –2.9 kcal/eq., respectively. If one equivalent of HA and one equivalent of HB are mixed with one equivalent of NaOH, the enthalpy change was found to be –6.9 kcal/eq., calculate the distribution of NaOH to neutralise HA and HB.

150. The standard enthalpy of formation of FeO and Fe_2O_3 is –65 kcal mol^{-1} and –197 kcal mol^{-1} respectively. If a mixture containing FeO and Fe_2O_3 in 2 : 1 mole ratio on oxidation is changed into 1 : 2 mole ratio, how much of thermal energy will be released per mole of the initial mixture?

151. A solution of 200 mL of 1 M KOH is added to 200 mL of 1 M HCl and the mixture after attaining reaction equilibrium shows a rise in temperature by ΔT_1. The experiment is repeated by using 50 mL of 1 M KOH and 50 mL 1 M HCl which at equilibrium shows a rise in temperature by ΔT_2. Thus:

(a) $\Delta T_1 = \Delta T_2$ (b) $\Delta T_1 = 4 \times \Delta T_2$

(c) $\Delta T_1 = 2 \times \Delta T_2$ (d) $\Delta T_1 > \Delta T_2$

152. Under the same conditions how many mL of 1 M KOH and 0.5 M H_2SO_4 solution respectively when mixed for a total volume of 100 mL, produce the highest rise in temperature?

(a) 67 : 33 (b) 33 : 67

(c) 40 : 60 (d) 50 : 50

153. The density of solid argon is 1.65 g/mL at –233°C. If the argon atom is assumed to be sphere of radius 1.54×10^{-8} cm, what percentage of solid argon is apparently empty space? (Atomic mass of Ar = 40)

154. A element crystallizes into a structure which may be described by a cubic type of unit cell having one atom on each corner of the cube and two atoms on one of its diagonals. If the volume of this unit cell is 24×10^{-24} cm^3 and density of element is 7.2 g cm^{-3}, calculate the number of atoms present in 200 g of element.

155. In diamond lattice, all lattice points and alternate tetrahedral voids are occupied by the carbon atoms. if diamond crystallizes in fcc form with edge length 'a', find out the:

(a) Number of next nearest neighbours in diamond lattice

(b) Distance between the next nearest neighbours

156. The γ-form of iron has fcc structure (edge length 386 pm) and β-form has bcc structure (edge length 290 pm). The ratio of density in γ-form and β-form is:

(a) 0.9788 (b) 1.02

(c) 1.57 (d) 0.6344

157. Two ionic solids AB and CB crystallise in the same lattice. If $\frac{r_{A^+}}{r_{B^-}}$ and $\frac{r_{C^+}}{r_{B^-}}$ are 0.50 and 0.70 respectively, then the ratio of edge length of AB and CB is:

(a) 0.68 (b) 0.78

(c) 0.88 (d) 0.98

158. Carbon, Silicon and Germanium have four valence electrons each. These are characterised by conduction bond separated by energy band gaps respectively equal to $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$. The correct statement is:

(a) $(E_g)_C = (E_g)_{Si} = (E_g)_{Ge}$ (b) $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$

(c) $(E_g)_C < (E_g)_{Ge} > (E_g)_{Si}$ (d) $(E_g)_{Si} < (E_g)_{Ge} > (E_g)_C$

❑ ❑ ❑

MATHEMATICS

1. Let ABC be a triangle. Let A be the point (1, 2), $y = x$ is the perpendicular bisector of AB and $x - 2y + 1 = 0$ is the angle bisector of angle C. If the equation of BC is given by $ax + by - 5 = 0$, then the value of $a + b$ is

   (1) 1 (2) 2

   (3) 3 (4) 4

2. Given a triangle whose vertices are at (0, 0), (4, 4) and (10, 0). A square is drawn in it such that its base is on the x-axis and its two corners are on the 2 sides of the triangle. The area of the square is equal to

   (1) $\frac{400}{49}$ (2) $\frac{400}{25}$

   (3) $\frac{625}{16}$ (4) $\frac{625}{49}$

3. The distance of any point (x, y) from the origin is defined as $d = \max\{|x|, |y|\}$, then the distance of the common point for the family of lines $x(1 + \lambda) + \lambda y + 2 + \lambda = 0$ (λ being parameter) from origin is

   (1) 1 (2) 2

   (3) $\sqrt{5}$ (4) 0

4. It is desired to construct a right angled triangle ABC ($\angle C = \pi/2$) in xy-plane so that its sides are parallel to co-ordinates axes and the medians through A and B lies on the lines $y = 3x + 1$ and $y = mx + 2$ respectively. The values of m for which such a triangle is possible is/are

   (1) –12 (2) 3/4

   (3) 4/3 (4) 1/12

5. A piece of cheese is located at (12, 10) in a coordinate plane. A mouse is at (4, –2) and is running up the line $y = -5x + 18$. At the point (a, b), the mouse starts getting farther from the cheese rather than closer to it. The value of $(a + b)$ is

   (1) 6 (2) 10

   (3) 18 (4) 14

6. Two sides of a rhombus $ABCD$ are parallel to lines $y = x + 2$ and $y = 7x + 3$. If the diagonals of the rhombus intersect at point (1, 2) and the vertex A is on the y-axis is, then the possible coordinates of A are

   (1) $\left(0, \frac{5}{2}\right)$ (2) (0, 0)

   (3) (0, 5) (4) (0, 3)

7. Line $\frac{x}{a}+\frac{y}{b}=1$ cuts the coordinate axes at $A(a, 0)$ and $B(0, b)$ and the line $\frac{x}{a'}+\frac{y}{b'}=-1$ at $A'(-a', 0)$ and $B'(0, -b')$. If the points A, B, A', B' are concyclic then the orthocentre of the triangle ABA' is

(1) $(0, 0)$ (2) $(0, b')$

(3) $\left(0, \frac{aa'}{b}\right)$ (4) $\left(0, \frac{bb'}{a}\right)$

8. Let $B(1, -3)$ and $D(0, 4)$ represent two vertices of rhombus $ABCD$ in (x, y) plane, then coordinates of vertex A if $\angle BAD = 60^\circ$ can be equal to

(1) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ (2) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

(3) $\left(\frac{-1+7\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}\right)$ (4) $\left(\frac{-1-7\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}\right)$

Paragraph for Q. Nos. 9 to 11

Consider a trapezoid $ABCD$, one of whose non parallel sides AB which is 8 cm long is perpendicular to the base. The base BC and AD of trapezoid are 6 cm and 10 cm in lengths respectively. Let L_1, L_2, L_3, L_4 represent the lines AB, BC, CD and DA respectively and $d(P, L)$ denote the perpendicular distance of point P from line L.

9. Find the area of region inside the trapezoid $ABCD$ in which the point Q can lie satisfying $d(Q, L_4) \le d(Q, L_3)$

(1) $3(3\sqrt{5}+\sqrt{3})$ (2) $24(\sqrt{3}-1)$

(3) $4(5-\sqrt{5})$ (4) $25(\sqrt{5}-1)$

10. Distance of the point R lying on line AD from vertex A so that perimeter of triangle RBC is minimum is

(1) 2 (2) 3

(3) 4 (4) 5

11. The maximum possible area of rectangle inscribed in the trapezoid so that one of its sides lies on the larger base of trapezoid is

(1) 36 (2) 54

(3) 42 (4) 48

Paragraph for Q. Nos. 12 to 14

Given point $A(6, 30)$ and point $B(24, 6)$, equation of line AB is $4x + 3y = 114$. Point $P(0, \lambda)$ is a point on y-axis such that $0 < \lambda < 38$ and point $Q(0, k)$ is a point on y-axis such that $k > 38$.

12. For all positions of point P, angle APB is maximum when point P is

(1) $(0, 12)$ (2) $(0, 15)$

(3) $(0, 18)$ (4) $(0, 21)$

13. The maximum value of angle APB is

(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{2}$

(3) $\frac{2\pi}{3}$ (4) $\frac{3\pi}{4}$

14. For all position of point Q, angle AQB is maximum when point Q is

(1) (0, 54) (2) (0, 58)

(3) (0, 60) (4) None of these

15. Let $D(0, \sqrt{3})$, $E(1, 0)$, $F(-1, 0)$ be the feet of perpendiculars dropped from vertices A, B, C to opposite sides BC, CA, AB respectively of triangle ABC.

Column-I	**Column-II**
(A) The ratio of the inradius of ΔABC to the inradius of ΔDEF is	(p) 2
(B) Let H be the orthocentre of ΔABC, then the greatest integer which is less than or equal to square of the length AH is	(q) 3
(C) The square of the sum of ordinates of points A, B and C is	(r) 4
(D) The length of side AB of ΔABC is	(s) 5

16. Given four parallel lines L_1, L_2, L_3 and L_4 as shown in figure. Let d_{ij} denote the perpendicular distance between lines L_i and L_j, $i, j \in \{1, 2, 3, 4\}$. Let P be a point, sum of whose perpendicular distances from four lines is K, also $d_{12} < d_{23} < d_{34}$. Then the complete locus of point P.

_____L_1

_____L_2

_____L_3

_____L_4

Column-I	**Column-II**
(A) If $K = d_{12} + 2d_{23} + d_{34}$	(p) Not possible
(B) If $K = d_{12} + 2d_{23} + d_{34} + 2\alpha$, where $0 < \alpha < d_{12}$	(q) Entire region between the lines L_2 and L_3
(C) If $K = d_{12} + 2d_{23} + d_{34} + 2\alpha$, where $0 < \alpha < d_{34}$	(r) Entire region between the lines L_1 and L_2
(D) If $K < d_{12} + 2d_{23} + d_{34}$	(s) Entire region between the lines L_1 and L_2 and between L_3 and L_4

17. $P(3, 1)$, $Q(6, 5)$ and $R(x, y)$ are three points such that angle PRQ is right angle and the area of ΔPRQ is 7, then number of such points R is

18. There exists two ordered triplets (a_1, b_1, c_1) and (a_2, b_2, c_2) for (a, b, c) for which the equation $4x^2 - 4xy + ay^2 + bx + cy + 1 = 0$ represents a pair of identical straight lines in x-y plane. Find the value of $a_1 + b_1 + c_1 + a_2 + b_2 + c_2$

19. The slopes of three sides of a triangle ABC are –1, –2, 3 respectively. If the orthocentre of triangle ABC is origin, then the locus of its centroid is $y = \frac{a}{b}x$ where a, b are relatively prime then $b - a$ is equal to

20. A variable line L_1 cuts $y = 3x + 1$ and $y = -2x + 3$ at points P_1 and P_2. If the locus of midpoints of P_1 and P_2 is line L_2 with undefined slope where slope of L_1 is constant. If slope of L_1 is $\frac{p}{q}$, where p, q are coprime natural number, then find $p + q$.

21. A circle with center (2, 2) touches the coordinate axes and a straight line AB where A and B lie on positive direction of coordinate axes such that the circle lies between origin and the line AB. If O be the origin then the locus of circumcenter of ΔOAB will be

(1) $xy = x + y + \sqrt{x^2 + y^2}$ (2) $xy = x + y - \sqrt{x^2 + y^2}$

(3) $xy + x + y = \sqrt{x^2 + y^2}$ (4) $xy + x + y + \sqrt{x^2 + y^2} = 0$

22. The value of α for which the point $(\alpha, \alpha + 2)$ is an interior point of smaller segment of the curve $x^2 + y^2 - 4 = 0$ made by the chord of the curve whose equation is $3x + 4y + 12 = 0$

(1) $\left(-\infty, -\frac{20}{7}\right)$ (2) $(-2, 0)$

(3) $\left(-\infty, \frac{20}{7}\right)$ (4) $\alpha \in \phi$

23. A circle is inscribed in an equilateral triangle with side lengths 6 unit. Another circle is drawn inside the triangle (but outside the first circle), tangent to the first circle and two of the sides of the triangle. The radius of the smaller circle is

(1) $1/\sqrt{3}$ (2) 2/3

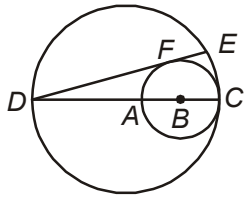
(3) 1/2 (4) 1

24. The centre of the smallest circle touching the circles $x^2 + y^2 - 2y - 3 = 0$ and $x^2 + y^2 - 8x - 18y + 93 = 0$ is

(1) (3, 2) (2) (4, 4)

(3) (2, 7) (4) (2, 5)

25. In the diagram, DC is a diameter of the large circle centered at A, and AC is a diameter of the smaller circle centered at B. If DE is tangent to the smaller circle at F and $DC = 12$ then the length of DE is



(1) $8\sqrt{2}$ (2) 16

(3) $9\sqrt{2}$ (4) $10\sqrt{2}$

26. The locus of the point of intersection of the tangent to the circle $x^2 + y^2 = a^2$, which include an angle of 45º is the curve $(x^2 + y^2)^2 = \lambda a^2(x^2 + y^2 - a^2)$. The value of λ is

(1) 2 (2) 4

(3) 8 (4) 16

27. Three concentric circles of which the biggest is $x^2 + y^2 = 1$, have their radii in A.P. If the line $y = x + 1$ cuts all the circles in real and distinct points. The interval in which the common difference of the A.P. will lie is

(1) $\left(0, \frac{1}{4}\right)$ (2) $\left(0, \frac{1}{2\sqrt{2}}\right)$

(3) $\left(0, \frac{2-\sqrt{2}}{4}\right)$ (4) None of these

28. *ABCD* is a square of unit area. A circle is tangent to two sides of *ABCD* and passes through exactly one of its vertices. The radius of the circle is

(1) $2-\sqrt{2}$ (2) $\sqrt{2}-1$

(3) $\frac{1}{2}$ (4) $\frac{1}{\sqrt{2}}$

29 A foot of the normal from the point (4, 3) to a circle is (2, 1) and a diameter of the circle has the equation $2x - y - 2 = 0$. Then the equation of the circle is

(1) $x^2+y^2-4y+2=0$ (2) $x^2+y^2-4y+1=0$

(3) $x^2+y^2-2x-1=0$ (4) $x^2+y^2-2x+1=0$

30. A circle of constant radius *a* passes through origin *O* and cuts the axes of coordinates in points *P* and *Q*, then the equation of the locus of the foot of perpendicular from *O* to *PQ* is

(1) $(x^2+y^2)\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=4a^2$

(2) $(x^2+y^2)^2\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=a^2$

(3) $(x^2+y^2)^2\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=4a^2$

(4) $(x^2+y^2)\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=a^2$

31. Two chords are drawn from the point $P(h, k)$ on the circle $x^2 + y^2 = hx + ky$. If the *y*-axis divides both the chords in the ratio 2 : 3, then which of the following may be correct?

(1) $k^2 > 15h^2$ (2) $15k^2 > h^2$

(3) $h^2 = 15k^2$ (4) $k^2 > 5h^2$

32. Let *x*, *y* be real variable satisfying the $x^2 + y^2 + 8x - 10y - 40 = 0$. Let $a = \max\left(\sqrt{(x+2)^2+(y-3)^2}\right)$ and $b = \min\left(\sqrt{(x+2)^2+(y-3)^2}\right)$, then

(1) $a + b = 18$ (2) $a+b=4\sqrt{2}$

(3) $a-b=4\sqrt{2}$ (4) $ab = 73$

33. Let l_1, l_2 and l_3 are the lengths of the tangents drawn from a variable point P to the circle $x^2 + y^2 = a^2$; $x^2 + y^2 = 2ax$ and $x^2 + y^2 = 2ay$ respectively. The lengths satisfy the relation $l_1^4 = l_2^2 l_3^2 + a^4$. Then the locus of P can be

(1) Line (2) Circle

(3) Parabola (4) Hyperbola

Paragraph for Q. Nos. 34 to 36

Consider two circles S_1 and S_2 (externally touching) having centres at points A and B whose radii are 1 and 2 respectively. A tangent to circle S_1 from point B intersects the circle S_1 at point C.D is chosen on circle S_2 so that AC is parallel to BD and the two segments BC and AD do not intersect. Segment AD intersect the circle S_1 at E. The line through B and E intersects the circle S_1 at another point F.

34. The length of segment EF is

(1) $2\sqrt{2}$ (2) $\frac{2\sqrt{2}}{3}$

(3) $\frac{2\sqrt{3}}{3}$ (4) $\sqrt{3}$

35. The area of triangle ABD is

(1) 2 (2) $\sqrt{3}$

(3) $2\sqrt{2}$ (4) $\sqrt{5}$

36. The length of the segment DE is

(1) $\sqrt{3}$ (2) $2\sqrt{2}$

(3) 2 (4) 3

Paragraph for Q. Nos. 37 to 39

Given a line segment AB, $A \equiv (0, 0)$ and $B(a, 0)$. Three circles S_1, S_2, S_3 of radius R are centered at the end points and the midpoint of the line segment AB. A fourth circles S_4 is drawn touching the 3 given circles.

37. If $0 < R < \frac{a}{4}$, then sum of all possible distinct values of radius of S_4 is

(1) $\frac{a^2}{16R}$ (2) $\frac{a^2}{7R}$

(3) $\frac{3a^2}{16R}$ (4) $\frac{a^2}{4R}$

38. If $0 < R < \frac{a}{4}$, then number of possible circles S_4 is

(1) 2 (2) 4

(3) 6 (4) 8

39. If $\frac{a}{4} < R < \frac{a}{2}$, then radius of circle S_4 is

(1) $\frac{a^2}{16R}$ (2) $\frac{a^2}{8R}$

(3) $\frac{3a^2}{16R}$ (4) $\frac{a^2}{4R}$

40. Let E, F, G, H be 4 distinct points inside square $ABCD$ whose area is 1 square units such that $\angle EDC = \angle ECD = \angle HDA = \angle HAD = \angle GAB = \angle GBA = \angle FBC = \angle FCB = 15^\circ$

Column-I	Column-II
(A) If area of quadrilateral $EFGH$ is equal to $a - \sqrt{b}$ where $a, b \in N$, then $a + b =$	(p) 1
(B) Let $\angle AEB = \frac{\pi}{k}$; then $k =$	(q) 3
(C) The radius of circle circumscribing the ΔAHD is equal to	(r) 5
(D) Let the lengths of perpendiculars from vertices G, A, H to opposite sides of triangle AHG be h_1, h_2, h_3 respectively. Let $\frac{1}{h_1^2} + \frac{1}{h_2^2} + \frac{1}{h_3^3} = a + \sqrt{b}$ where $a, b \in N$, then $\frac{b}{a} =$	(s) 6

41. In the parallelogram $ABCD$ with angle $A = 60^\circ$, the bisector of angle B is drawn which cuts the side CD at a point E. A circle S_1 of radius r is inscribed in the ΔECB. Another circle S_2 is inscribed in the trapezoid $ABED$.

Column-I	Column-II
(A) The value of radius of S_2 is	(p) $2\sqrt{3}r$
(B) The value of distance between the centres of S_1 and S_2 is	(q) $\frac{\sqrt{3}}{2}r$
(C) The value of the length of internal common tangent of S_1 and S_2 is	(r) $\sqrt{7}\,r$
(D) The value of the length CE is	(s) $\frac{3}{2}r$

42. Triangles *ABC* are described on a given base *BC* and of a given vertical angle α.

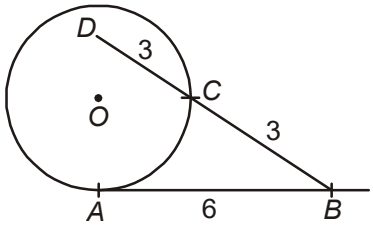
Column-I	Column-II
(A) The locus of orthocentre of ΔABC is	(p) Part of circle such that the *BC* subtend angle $\pi - \alpha$ its circumference
(B) The locus of incentre of ΔABC is	(q) Part of circle such that *BC* subtend angle $\frac{\pi}{2}+\frac{\alpha}{2}$ at its circumference
(C) The locus of the excentre corresponding to vertex opposite to base of ΔABC is	(r) Part of circle such that *BC* subtend angle $\frac{\pi}{2}-\frac{\alpha}{2}$ at its circumference
(D) The locus of centroid of ΔABC is	(s) Part of circle such that *PQ* subtend angle α at its circumference where *P*, *Q* are points of trisection of segment *BC*

43. Let $S = \{(x, y) : x^2 + y^2 - 6x - 8y + 21 \le 0\}$

Match the Column-I with Column-II.

Column-I	Column-II
(A) $\max\left\{\frac{12x}{7}-\frac{5y}{7};\ (x, y)\in S\right\}$	(p) 3
(B) $\min\left\{\frac{1}{2}(x^2+y^2+1)+(x-y);\ (x, y)\in S\right\}$	(q) 4
(C) $\max\left\{\frac{3x}{7}+\frac{4y}{7};\ (x, y)\in S\right\}$	(r) 5
(D) $\min\left\{\frac{\sqrt{3}y+\lvert x-3\rvert}{\lvert x-3\rvert};\ (x, y)\in S\right\}$	(s) 6
	(t) 7

44. In the figure *AB* is tangent at *A* to circle with centre *O*; point *D* is interior to circle and *DB* intersects the circle at *C*. If *BC* = *DC* = 3, *OD* = 2 and *AB* = 6, then find the radius of the cricle.



45. Two circle of radii 8 and 4 touch each other externally at a point *A*. Through a point *B* taken on the larger circle a straight line is drawn touching the smaller circle at *C*. Find *BC* if $AB = \sqrt{6}$.

46. Parabolas $(y-\alpha)^2 = 4a(x-\beta)$ and $(y-\alpha)^2 = 4a'(x-\beta')$ will have a common normal (other than the normal passing through vertex of parabola) if

(1) $\frac{2(a-a')}{\beta'-\beta} < 1$ (2) $\frac{2(a-a')}{\beta-\beta'} < 1$

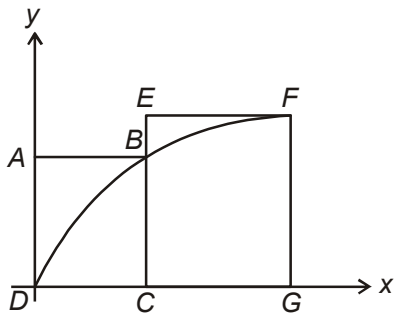
(3) $\frac{2(a'-a)}{\beta+\beta'} < 1$ (4) $\frac{2(a'-a)}{\beta+\beta'} > 1$

47. If the line $x + y - 1 = 0$ is a tangent to a parabola with focus (1, 2) at *A* and intersects the directrix at *B* and tangent at vertex at *C* respectively, then *AC* . *BC* is equal to

(1) 2 (2) 1

(3) $\frac{1}{2}$ (4) $\frac{1}{4}$

48. *ABCD* and *EFGC* are squares and the curve $y = k\sqrt{x}$ passes through the origin *D* and the points *B* and *F*. The ratio $\frac{FG}{BC}$ is



(1) $\frac{\sqrt{5}+1}{2}$ (2) $\frac{\sqrt{3}+1}{2}$

(3) $\frac{\sqrt{5}+1}{4}$ (4) $\frac{\sqrt{3}+1}{4}$

49. A parabola $y = ax^2 + bx + c$ crosses the *x*-axis at $(\alpha, 0)$ $(\beta, 0)$ both to the right of the origin. A circle also passes through these two points. The length of a tangent from the origin to the circle is

(1) $\sqrt{\frac{bc}{a}}$ (2) ac^2

(3) $\frac{b}{a}$ (4) $\sqrt{\frac{c}{a}}$

50. The locus of the foot of the perpendiculars drawn from the vertex on a variable tangent to the parabola $y^2 = 4ax$ is

(1) $x(x^2+y^2)+ay^2=0$ (2) $y(x^2+y^2)+ax^2=0$

(3) $x(x^2-y^2)+ay^2=0$ (4) None of these

51. In a square matrix A of order 3, $a_{ii} = m_i + i$ where $i = 1, 2, 3$ and m_i's are the slopes (in increasing order of their absolute value) of the 3 normals concurrent at the point (9, –6) to the parabola $y^2 = 4x$. Rest all other entries of the matrix are one. The value of det. (A) is equal to

(1) 37 (2) –6

(3) –4 (4) –9

52. A variable chord PQ of the parabola $y^2 = 4ax$ is drawn parallel to the line $y = x$. If the parameter of the points P and Q on the parabola be t_1 and t_2 respectively, then

(1) $t_1 + t_2 = 2$

(2) $t_1 t_2 = \frac{2}{a}$

(3) Locus of point of intersection of tangents at P and Q is $y = 2a$

(4) Locus of point of intersection of normals at P and Q is $2x - y = 12a$

53. If P_1P_2 and Q_1Q_2, two focal chords of a parabola are at right angles, then

(1) Area of the quadrilateral $P_1Q_1P_2Q_2$ is minimum when the chords are inclined at an angle $\frac{\pi}{4}$ to the axis of the parabola

(2) Minimum area is twice the area of the square on the latus rectum of the parabola

(3) Minimum area of $P_1Q_1P_2Q_2$ cannot be found

(4) Minimum area is thrice the area of the square on the latus rectum of the parabola

Paragraph for Q. Nos. 54 to 56

$y = f(x)$ is a parabola of the form $f(x) = x^2 + bx + 1$, b is a constant. The tangent line is drawn at the point where $f(x)$ cuts y-axis, also touches $x^2 + y^2 = r^2 (r > 0)$. It is also given that at least one tangent can be drawn from point P to $y = f(x)$ where P is a point at which $y = |x - \alpha|$ is non differentiable $\forall\ \alpha \in R$.

54. For maximum value of b, the area of circle is

(1) $\frac{\pi}{10}$ (2) $\frac{\pi}{5}$

(3) π (4) 5π

55. $\lim\limits_{b \to 0} \frac{\sqrt{r_{max.} - r}}{\sin b} =$

(1) $\frac{1}{\sqrt{2}}$ (2) $-\frac{1}{\sqrt{2}}$

(3) $\frac{1}{2}$ (4) Non exist

56. Locus of vertex of parabola is

(1) $y = 1 - x^2$, $x \in [-1, 1]$, $y \in [-1, 0]$

(2) $y = 1 - x^2$, $x \in [-2, 2]$, $y \in [0, 1]$

(3) $y = 1 - x^2$, $x \in [-2, 2]$, $y \in [-3, 1]$

(4) $y = 1 - x^2$, $x \in [-1, 1]$, $y \in [0, 1]$

Paragraph for Q. Nos. 57 to 59

Let curve S_1 be the locus of a point $P(h, k)$ which moves in xy plane such that it always satisfy the relation $\min\{x^2 + (h-k)x + (1-h-k)\} = \max\{-x^2 + (h+k)x - (1+h+k)\}$. Let S_2 is a curved mirror passing through (8, 6) having the property that all light says emerging from origin, after getting reflected from the mirror becomes parallel to x-axis. Also the area of region bounded between y-axis and S_2 is 8/3.

57. The area of smaller region bounded between S_1 and S_2 is equal to

(1) 2π (2) $\pi - \frac{8}{3}$

(3) $\pi + \frac{8}{3}$ (4) $2\pi - \frac{8}{3}$

58. If the circle $(x-4)^2 + y^2 = r^2$ internally touches the curve S_2, then r =

(1) 5 (2) 4

(3) 3 (4) 2

59. The ratio in which the curve $y = \left[\sin^2\frac{x}{4} + \cos\frac{x}{4}\right]$, where [.] denote greatest integer function divides the curve S_1 is

(1) $4\pi + 3\sqrt{3} : 8\pi - 3\sqrt{3}$ (2) $4\pi - 3\sqrt{3} : 8\pi + 3\sqrt{3}$

(3) 1 : 1 (4) $4\pi - \sqrt{3} : 8\pi + \sqrt{3}$

60. If $y = x + 1$ is axis of parabola, $y + x = 4$ is tangent of same parabola at its vertex and $y = 2x + 3$ is one of its tangent, then

Column-I	**Column-II**
(A) If equation of directrix of parabola is $ax + by - 29 = 0$, then $a + b$ =	(p) 9
(B) If length of latus rectum of parabola is $\frac{a\sqrt{2}}{b}$ where a and b are relatively prime natural numbers, then $a + b$ =	(q) 18
(C) Let extremities of latus rectum are (a_1, b_1) and (a_2, b_2), then $[a_1 + b_1 + a_2 + b_2]$ = (where [.] denote greatest integer function)	(r) 23
(D) If equation of parabola is $a(x-y+1)^2 = b(x+y-4)$ where a and b are relatively prime natural numbers then $a + b$ =	(s) 37

61. Normals are drawn from point (4, 1) to the parabola $y^2 = 4x$. The tangents at the feet of normals to the parabola $y^2 = 4x$ form a triangle ABC.

Column-I	Column-II
(A) The distance of focus of parabola $y^2 = 4x$ from centroid of ΔABC is	(p) $\frac{5}{3}$
(B) The distance of focus of parabola $y^2 = 4x$ from orthocentre of ΔABC is	(q) $\frac{\sqrt{10}}{2}$
(C) The distance of focus of parabola $y^2 = 4x$ from circumcentre of ΔABC is	(r) $\frac{\sqrt{7}}{2}$
(D) Area of ΔABC is	(s) $\frac{\sqrt{5}}{2}$
	(t) $\sqrt{5}$

62. If the focus of parabola $y^2 + 8 = 4x$ coincides with one of the foci of ellipse $3x^2 + by^2 - 12x = 0$, then the reciprocal of eccentricity of ellipse is

63. A series of ellipses $E_1, E_2, \ldots E_n$ are drawn such that E_n touches E_{n-1} at the extremeties of the major axis of E_{n-1} and the foci of E_n coincide with the extremeties of minor axis of E_{n-1}. If ecentricity of the ellipses is independent of n, then the value of ecentricity is

(1) $\frac{\sqrt{5}}{3}$ (2) $\frac{\sqrt{5}+1}{4}$

(3) $\frac{\sqrt{5}-1}{2}$ (4) $\frac{\sqrt{5}-1}{3}$

64. If normal at any point P to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\,(a>b)$ meet the axes at M and N so that $\frac{PM}{PN}=\frac{2}{3}$, then the value of eccentricity is

(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{\sqrt{2}}{\sqrt{3}}$

(3) $\frac{1}{\sqrt{3}}$ (4) None of these

65. If S and S' are the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and P is any point on it, then difference of maximum and minimum of $SP.S'P$ is equal to

(1) 16 (2) 9

(3) 15 (4) 25

66. The area of the rectangle formed by the perpendiculars from the centre of the standard ellipse to the tangent and normal at its point whose eccentric angle is $\pi/4$, is

(1) $\frac{(a^2-b^2)ab}{a^2+b^2}$ (2) $\frac{(a^2-b^2)}{(a^2+b^2)ab}$

(3) $\frac{(a^2-b^2)}{ab(a^2+b^2)}$ (4) $\frac{a^2+b^2}{(a^2-b^2)ab}$

67. *P* and *Q* are two points on the upper half of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. The centre of the ellipse is at the origin *O* and *PQ* is parallel to the *x*-axis such that the triangle *OPQ* has the maximum possible area. A point is randomly selected from inside of the upper half of the ellipse. The probability that is lies outside the triangle, is

(1) $\frac{\pi-1}{\pi}$ (2) $\frac{2\pi-1}{2\pi}$

(3) $\frac{\pi-1}{2\pi}$ (4) $\frac{\pi-1}{4\pi}$

68. Line through *P*(*a*, 2) meets the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ at *A* and *D* and meets the coordinate axes at *B* and *C* so that *PA*, *PB*, *PC*, *PD* are in G.P., then possible values of *a* can be

(1) 5 (2) 8

(3) 10 (4) –7

69. A parallelogram circumscribes the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and two of its opposite angular points lie on straight lines $x^2 = \lambda^2$, $\lambda \neq 0$, the locus of the other two vertices is

(1) Ellipse if $\lambda > 3$ (2) Hyperbola if $\lambda \in (0, 3)$

(3) Circle if $\lambda = \frac{9}{\sqrt{5}}$ (4) Ellipse if $\lambda \in (0, 3)$

Paragraph for Q. Nos. 70 to 72

An ellipse has major axis of length 4 and minor axis of length 2. It is slipping between the coordinate axes in the first quadrant while maintaining contact with both *x*-axis and *y*-axis.

70. The locus of the centre of the ellipse is

(1) $x + y = 3$ (2) $(y-1)^2 = 4(x-2)$

(3) $x^2 + y^2 = 5$ (4) $x^2 + y^2 = 20$

71. The locus of foci of the ellipse is

(1) $x^2 + y^2 = 10$ (2) $x^4 + y^4 - x^2y^2 = 1$

(3) $x^2 + y^2 (x^2 + y^2) = 4$ (4) $x^2+y^2+\frac{1}{x^2}+\frac{1}{y^2}=16$

72. When the ordinate of centre of ellipse is 1, a light ray passing through origin and centre of ellipse strikes the tangent at vertex of the parabola $y^2 + 4x - 10y - 15 = 0$. The reflected ray cuts y-axis at the point.

(1) (0, 8) (2) (0, 10)

(3) (0, 12) (4) (0, 16)

73. Rectangle $ABCD$ has area 200. An ellipse with area 200 π passes through A and C and has foci at B and D. Let perimeter of the rectangle $ABCD$ is P, then $\frac{P}{10}$ =

74. If the equation of the curve on reflection of the ellipse $\frac{(x-4)^2}{16}+\frac{(y-3)^2}{9}=1$ about the line $x - y - 2 = 0$ is $16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$, then $\frac{k_1+k_2}{22}$ is _____.

75. A circle concentric to the ellipse $\frac{4x^2}{289}+\frac{y^2}{\lambda^2}=1\left(\lambda<\frac{17}{2}\right)$ passes through foci S_1, S_2 and cuts ellipse at point P. If area of ΔPS_1S_2 is 30 sq. units then find $\frac{S_1S_2}{13}$.

76. If the tangent at the point $P(h, k)$ on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ cuts the circle $x^2 + y^2 = a^2$ at the points $Q(x_1, y_1)$ and $R(x_2, y_2)$ then $\frac{1}{y_1}+\frac{1}{y_2}$ is equal to

(1) $\frac{2}{k}$ (2) $\frac{1}{k}$

(3) $\frac{a}{k}$ (4) $\frac{b}{k}$

77. Let the major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to $2R$ and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola then the correct relation is

(1) $4e_1^2 - e_2^2 = 6$ (2) $e_1^2 - 4e_2^2 = 2$

(3) $4e_2^2 - e_1^2 = 6$ (4) $2e_1^3 - e_2^2 = 4$

78. A conic passes through the point (2, 4) and is such that the segment of any of its tangents at any point contained between the co-ordinate is bisected at the point of tangency. Then the foci of the conic are

(1) $(2\sqrt{2}, 0)$ and $(-2\sqrt{2}, 0)$ (2) $(2\sqrt{2}, 2\sqrt{2})$ and $(-2\sqrt{2}, -2\sqrt{2})$

(3) (4, 4) and (–4, –4) (4) $(4\sqrt{2}, 4\sqrt{2})$ and $(-4\sqrt{2}, -4\sqrt{2})$

79. Number of common tangent with finite slope to the curves $xy = c^2$ and $y^2 = 4ax$ is

(1) 0 (2) 1

(3) 2 (4) 4

80. The normal at one extremity of latus rectum (in 1st quadrant) of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,\ a>b>0$ meets the rectangular hyperbola $xy = 9$ at points P and Q, then

(1) If P is $\left(6, \frac{3}{2}\right) \Rightarrow Q$ is $\left(-\frac{3\sqrt{2}}{2}, -3\sqrt{2}\right)$

(2) Eccentricity of hyperbola is $\sqrt{2}$

(3) If P is $\left(6, \frac{3}{2}\right) \Rightarrow Q$ is $\left(-\frac{3e}{2}, -\frac{6}{e}\right)$ where e is eccentricity of the given ellipse

(4) If O is origin, then product of slopes of OP and OQ is positive

81. The differential equation $\frac{dx}{dy}=\frac{3y}{2x}$ represents a family of hyperbolas (except when it represents a pair of lines) with eccentricity

(1) $\sqrt{\frac{3}{5}}$ (2) $\sqrt{\frac{5}{3}}$

(3) $\sqrt{\frac{2}{5}}$ (4) $\sqrt{\frac{5}{2}}$

Paragraph for Q. Nos. 82 to 84

Consider an ellipse $\frac{x^2}{36}+\frac{y^2}{18}=1$. There is a hyperbola whose one asymptotes is major axis of given ellipse. If eccentricity of given ellipse and hyperbola are reciprocal to each other, both have same centre and both touch each other in first and third quadrant.

82. Focus of hyperbola is equal to

(1) $\left(\frac{3}{2}, \frac{3}{2}\right)$ (2) $\left(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$

(3) $(3\sqrt{2}, 3\sqrt{2})$ (4) $\left(3(2^{3/4}), 3(2^{3/4})\right)$

83. Number of points in x-y plane from where $\perp$ tangents can be drawn to hyperbola

(1) Zero (2) 1

(3) Infinite (4) None of these

84. The equation of common tangent to given ellipse and hyperbola in first quadrant is

(1) $\frac{x}{\sqrt{2}}+y=3$ (2) $\frac{x}{\sqrt{2}}+y=6$

(3) $x+y\sqrt{2}=3\sqrt{2}$ (4) $x+y\sqrt{2}=6$

85. Let the foci of the hyperbola $\frac{x^2}{A^2}-\frac{y^2}{B^2}=1$ be the vertices of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the foci of the ellipse be the vertices of the hyperbola. Let the eccentricities of the ellipse and hyperbola be e_1 and e_2 respectively.

Column-I	**Column-II**
(A) $\frac{b}{B}$	(p) 1
(B) $e_1 + e_2$ cannot be equal to	(q) 2
(C) If the angle between the asymptotes of the hyperbola is $\frac{2\pi}{3}$, then $2e_1 \le$	(r) 3
(D) If $e_1=\frac{1}{\sqrt{2}}$ and (x, y) is point of intersection of ellipse and hyperbola, then $\frac{x^2}{y^2}=$	(s) 4

86. The tangent to a curve at a point $P(x, y)$ meets the x-axis at T and y-axis at S while the normal at P meets the x-axis at N and y-axis at M, O is the origin. Match the locus of point P satisfying the condition in Column-I to the curve given in Column-II.

Column-I	**Column-II**
(A) $TP = PS$	(p) Straight line
(B) $NM = NP$	(q) Circle
(C) $TP = OP$	(r) Ellipse
(D) $NP = OP$	(s) Hyperbola

87. The graphs of $x^2 + y^2 + 6x - 24y + 72 = 0$ and $x^2 - y^2 + 6x + 16y - 46 = 0$, intersect at 4 points. Compute the sum of distances of the 4 points from point $(-3, 2)$.

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